Notes for Complex Analysis

Made by Gene

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0 Reading Week

0.1 Algebra of Complex Numbers

Complex numbers are an extension of $\mathbb{R}$. We adjoin $\sqrt{-1}$ and allow addition & multiplication. The set of complex numbers is denoted $\mathbb{C}$.

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

For every complex number $z \in \mathbb{C}$, the real part is denoted $\mathrm{Re}(z)$ and the imaginary part is denoted $\mathrm{Im}(z)$.

$$z = a + bi \longrightarrow \begin{cases} \mathrm{Re}(z) = a \\ \mathrm{Im}(z) = b \end{cases}$$

We denote the absolute value or "modulus" of z as $|z|$. It's pretty simple, $|z|^2 = a^2 + b^2$ for $z = a + bi$.

$$|z|^2 = (\mathrm{Re}(z))^2 + (\mathrm{Im}(z))^2$$

The Inverse: For every $z \in \mathbb{C}$ that is nonzero, there exists a $z^{-1} \in \mathbb{C}$ such that $zz^{-1} = 1$. Namely,

$$z^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i$$

The Complex Conjugate: For every $z = a + bi \in \mathbb{C}$, the complex conjugate is $\bar{z} = a - bi$. Addition, subtraction, multiplication, and division work as expected, which is nice.

$$\begin{gathered} |z|^2 = z\overline{z} \\ \overline{z_1 + z_2} = \overline{z_1} + \overline{z_2} \\ \overline{z_1 - z_2} = \overline{z_1} - \overline{z_2} \\ \overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2} \\ \overline{z_1 / z_2} = \overline{z_1} / \overline{z_2} \end{gathered}$$

The Triangle Inequalities: Cf. triangle inequalities for $\mathbb{R}$

$$\begin{gathered} |z_1 + z_2| \leq |z_1| + |z_2| \\ |z_1 - z_2| \geq \Big| |z_1| - |z_2| \Big| \end{gathered}$$

0.2 Polar Coordinates

Let $z = r(\cos\theta + i\sin\theta)$. Then there are multiple values of θ that can create a given z (e.g. θ, $\theta + 2\pi$, $\ldots$)

A possible θ for a given z is called an argument of z, denoted $\arg(z)$.

$$\arg(z) = \{\theta \mid z = r(\cos\theta + i\sin\theta)\}$$

A unique argument θ of z satisfying $-\pi < \theta \leq \pi$ is called the principal argument of z, denoted Arg(z). It can be thought of as a function from $\mathbb{C}$ (excluding 0) to a number between $-\pi$ and π.

$$\text{Arg} : \mathbb{C} \setminus \{0\} \longrightarrow (-\pi, \pi]$$

We can represent a complex number $z \in \mathbb{C}$ in polar coordinates, using $r = |z|$ and $\theta \in \arg(z)$. It's often easier to multiply and divide complex numbers in this form.

$$z = re^{i\theta}$$

1 Week 1 (Jan 12 – 16)

1.1 Roots of Complex Numbers

- All nonzero real numbers have 2 square roots in $\mathbb{C}$.
- All nonzero elements of $\mathbb{C}$ have a square root in $\mathbb{C}$ (cf: not the case in $\mathbb{R}$).
- Note that 0 has only 1 square root: 0.

For any positive, nonzero integer n and $z \in \mathbb{C}$, all n-th roots of z lie in $\mathbb{C}$. The set of n-th roots of $z = re^{i\theta}$ are given by

$$\left\{ \sqrt[n]{r}e^{i\left(\frac{\theta+2\pi k}{n}\right)} \mid k = 0, 1, 2, ..., n-1 \right\}$$

Proof: Let $a_k = \sqrt[n]{r}e^{i(\theta+2\pi k)/n}$. Then a calculation shows that $a_k^n = re^{i\theta} = z$ as expected.

We also need to show that all $a_0, a_1, ..., a_{n-1}$ are distinct. Proof by contradiction: assume $a_r = a_s$ and $0 \leq r < s \leq n-1$. Then $(\theta + 2\pi r)/n = (\theta + 2\pi s)/n - 2\pi j$ for some integer j. Solving, we should get

$$\frac{2\pi(r-s)}{n} = 2\pi j$$

But this is impossible, since $s - r < n$. This is a contradiction, so all $a_0, a_1, ..., a_{n-1}$ must be distinct.

We know that a polynomial of degree n has at most n zeros. Apply this to $x^n - z$ to see that z has at most n n-th roots. So $a_0, a_1, ..., a_k$ are all the n-th roots of z.

Geometrically, if $n > 0$ and $z \neq 0$, then the elements of the set $z^{1/n}$ lie at the vertices of a regular n-gon centered at the origin. Each vertex is equidistant from 0, and the angle between successive vertices is $2\pi/n$.

A special case is the n-th roots of 1, or "unity." These are given by

$$\left\{ e^{i0}, e^{2\pi i/n}, e^{4\pi i/n} ..., e^{2(n-1)\pi i/n} \right\}$$

If $\omega^n = 1$ but $\omega^k \neq 1$ if $k < n$, then ω is a *primitive n-th root of unity*. For example, let $\omega = e^{2\pi i/n}$. $\omega^n = 1$, but not for any $0 < k < n$.

We can define the set of n-th roots using primitive n-th roots. Let $z \neq 0$, $\alpha^n = z$, and ω be a primitive n-th root of unity. Then

$$z^{1/n} = \left\{\alpha, \alpha\omega, \alpha\omega^2, ..., \alpha\omega^{n-1}\right\}$$

Proof: For any k, $(\alpha\omega^k)^n = z(1^k) = z$, so $\alpha\omega^k$ is indeed an n-th root.

The n-th roots are also unique. To prove this, assume $\alpha\omega^s = \alpha\omega^r$ for $0 \leq r < s \leq n-1$. Then $\omega^{s-r} = 1$, but $0 < s - r < n$, so ω can't be a primitive n-th root of unity.

1.2 Open vs. Closed Sets

Given $z_0 \in \mathbb{C}$ and $\epsilon \in \mathbb{R}$, we define the ε-neighborhood of z_0 as the subset

$$\{z \mid |z - z_0| < \varepsilon\}$$

We also define the deleted ε-neighborhood as the subset

$$\{z \mid 0 < |z - z_0| < \varepsilon\}$$

The ε-neighborhood is like a circular region around z_0, except the outline. The deleted ε-neighborhood is that minus z_0.

We call $z_0 \in \mathbb{C}$ as an interior point of some set S if there exists an ε-neighborhood of z_0 contained in S.

Similarly, $z_0 \in \mathbb{C}$ is an exterior point of some set S if there exists an ε-neighborhood of z_0 contained in the complement of S, namely $\mathbb{C} \setminus S = \{z \in \mathbb{C} \mid z \notin S\}$.

There also exists points that are neither interior nor exterior points of S. This is the boundary of S, denoted ∂S. A point z_0 is in ∂S iff. every ε-neighborhood of z_0 intersects both S and $\mathbb{C} \setminus S$.

A set S is open if no boundary points lie in S (think of intervals from Calc BC).

$$S \cap \partial S = \emptyset$$

Theorem: S is open iff. every point in S is an interior point.

Proof: If $S = \emptyset$, there's nothing to prove, so assume $S \neq \emptyset$.

- First direction: Let $z \in S$. If S is open, then $z \notin \partial S$, but since $z \in S$, z must be an interior point.
- Second direction: If every $z \in S$ is an interior point, then for every z, we have $z \notin \partial S$. Use the notation $S = \{z\}$, the set of all z. Then $S \cap \partial S = \emptyset$, so S must be open.

A set S is closed if its boundary lies in S (again, think of closed intervals from Calc BC).

$$\partial S \subset S$$

For example, $\{z \mid |z-1| \leq 2\}$ has the boundary given by $|z-1| = 2$.

For any given S, the set $S \cup \partial S$ is closed. We call $S \cup \partial S$ the *closure* of S. The closure of S is the intersection of all closed sets containing S.

Note 1: A set can be neither open nor closed. Back to the Calc BC example, think of the interval $[0, 1)$. For sets, it's similar. A set could contain just a few of its boundary points.

Note 2: If S is both open and closed, $S = \mathbb{C}$ or $S = \emptyset$.

We call S connected if for $z_1, z_2 \in S$, there exists a polygonal path in S, composed of straight line segments, that connects z_1 and z_2.

S is bounded if for some $r \in \mathbb{R}$, S can be constrained in a finite disk:

$$S \subset \{z \mid |z| < r\}$$

We call z_0 an accumulation point of S if every deleted ε-neighborhood of z_0 contains points of S (except z_0, per definition). It's kind of like the "center" of a set, I guess.

Theorem: If S is closed, then S contains all its accumulation points.

Proof: Suppose z is an accumulation point of a closed set S, and $z \notin S$. Then each ε-neighborhood of z contains a point not in S. But every ε-neighborhood of z must contain some point in S, since z is an accumulation point of S. Every ε-neighborhood of z contains both points in S and points not in S, thus $z \in \partial S$.

Since S is closed, $\partial S \subset S$, so $z \in S$. This contradicts our assumption that $z \notin S$, which completes the proof.

A domain is defined to be a set that's both open and connected (think of intervals on the real line in Calc). This can conflict with our usual definition of "domain" as the inputs to a function, but they're different. Be careful.

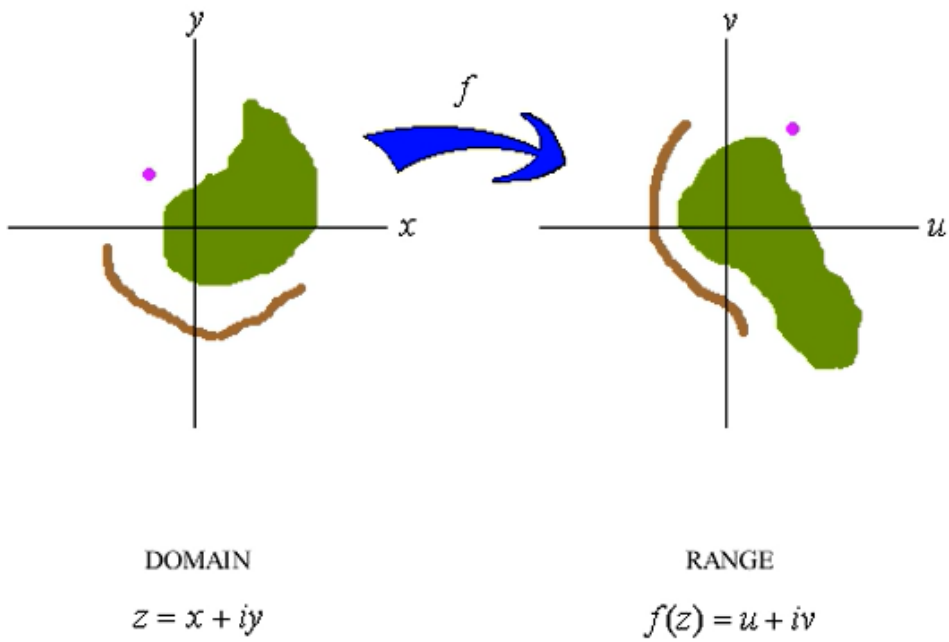
A region is a domain plus some (maybe none, maybe all) of its boundary points.

1.3 Complex Functions

A function f defined on set A is a rule that assigns to each $a \in A$ an element $f(b) \in B$.

In Complex Analysis, we'll consider functions of a complex value: $A \subset C$.

Graphing is a bit tricky because we have to map from a complex plane $\longrightarrow$ complex plane, total 4 dimensions. Use colors to get around this.



Graphing Complex-Valued Functions. Consider points, lines, or curves in the domain, parametrize them, and map with the function.

2 Week 2 (Jan 19 – 23)

2.1 Limits of Complex Functions

Suppose f (function of a complex variable) is defined in some deleted ϵ-neighborhood of z_0, and w_0 is some complex number. Then $\lim_{z \to z_0} f(z) = w_0$ means

For every $\epsilon > 0$, there exists $\delta > 0$ such that for every z in the domain of f, $0 < |z - z_0| < \delta$ implies $|f(z) - w_0| < \epsilon$

An equivalent formulation can be made in terms of neighborhoods:

For any ϵ-neighborhood of w_0, there exists a deleted δ-neighborhood of z_0 such that f takes every point in this deleted δ-neighborhood into the ϵ-neighborhood of w_0.

Baically, as the input z gets arbitrarily close to z_0, the function's output gets arbitrarily close to w_0. If that happens, w_0 is the limit.

Limit theorems are pretty much the same from Real Analysis. Suppose that $\lim_{z \to z_0} f_1(z) = w_1$ and $\lim_{z \to z_0} f_2(z) = w_2$. Then

i) Limits are unique. If for some $w \in \mathbb{C}$, we have $\lim_{z \to z_0} f_1(z) = w$, then $w = w_1$.

ii) $\lim_{z \to z_0} (f_1 + f_2)(z) = w_1 + w_2$

iii) $\lim_{z\to z_0}(cf_1)(z) = cw_1$

iv) $\lim_{z\to z_0}(f_1 \cdot f_2)(z) = w_1 w_2$

v) $\lim_{z\to z_0}\left(\frac{f_1}{f_2}\right)(z) = \frac{w_1}{w_2}$, given $w_2 \neq 0$

We can also reduce complex-valued limits to real-valued ones.

Theorem: Let $f(z) = u(x,y) + iv(x,y)$, $z_0 = x_0 + iy_0$, and $w_0 = u_0 + iv_0$. Then

$$\lim_{z\to z_0} f(z) = w_0 \text{ if and only if}$$
$$\lim_{(x,y)\to(x_0,y_0)} u = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v = v_0$$

Limits of polynomials are similar to before. Given $P(z) = c_0 + c_1 z + \cdots + c_n z^n$,

$$\lim_{z\to z_0} P(z) = P(z_0)$$

If $\lim_{z\to z_0} f(z)$ exists, then it can be computed by considering f restricted to *any* path of inputs that get arbitrarily close to z_0. The specific path doesn't matter.

For example, when we try to find the limit $\lim_{z\to 0} f(z)$ for $f(z) = \text{Re}(z)/z$, we can consider two paths:

1. Path such that $\text{Re}(z) = z$ (pure real), and z gets arbitrarily close to 0.
2. Path such that $\text{Re}(z) = 0$ (pure imaginary), and z gets arbitrarily close to 0.

In the first case, $\lim_{z\to 0} f(z) = 1$, but in the second case, $\lim_{z\to 0} f(z) = 0$. Thus the limit of f doesn't exist at $z = 0$.

2.2 Limits Involving ∞

For complex numbers, there's no distinction between $-\infty$ and $+\infty$. When we say $z \to \infty$, we just mean that z gets arbitrarily far away from 0, so $|z| \to +\infty$.

There are three basic cases for limits:

i) $\lim_{z\to z_0} f(z) = \infty$

For every $\epsilon > 0$, there exists $\delta > 0$ such that for every z in the domain of f, $0 < \lvert z - z_0\rvert < \delta$ implies $\lvert f(z)\rvert > \frac{1}{\epsilon}$

ii) $\lim_{z\to\infty} f(z) = w$

For every $\epsilon > 0$, there exists $\delta > 0$ such that for every z in the domain of f, $\lvert z\rvert > \frac{1}{\delta}$ implies $\lvert f(z) - w\rvert < \epsilon$

iii) $\lim_{z\to\infty} f(z) = \infty$

For every $\epsilon > 0$, there exists $\delta > 0$ such that for every z in the domain of f, $\lvert z\rvert > \frac{1}{\delta}$ implies $\lvert f(z)\rvert > \frac{1}{\epsilon}$

Note that $\infty \notin \mathbb{C}$. But it's a useful concept to use. We can rewrite limits involving ∞ with real numbers by taking the reciprocal. So for example, limit (i) and (ii) could be rewritten as

$$\lim_{z\to z_0} f(z) = \infty \longrightarrow \lim_{z\to z_0} \frac{1}{f(z)} = 0$$
$$\lim_{z\to\infty} f(z) = w \longrightarrow \lim_{z\to 0} f\left(\frac{1}{z}\right) = w$$

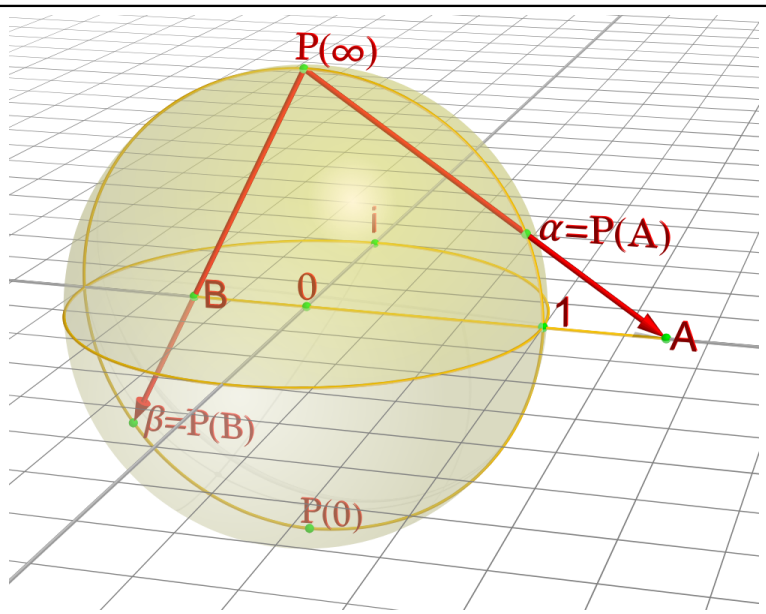
Example: Show that $\lim_{z\to\infty} f(z) = \infty$, where $f(z) = \frac{1+z+iz^4}{z^2-2}$.

Rewriting, we want to prove that

$$\lim_{z\to 0} \frac{1}{f(1/z)} = 0$$

which is easily seen by substituting and using the appropriate limit theorems.

An interpretation of ∞ can be done with a Riemann sphere.



https://upload.wikimedia.org/wikipedia/commons/3/32/Riemann_sphere1.svg

The plane in which the sphere lies is the complex plane. The North Pole of the sphere is ∞. We see that all points in the plane correspond to points on the sphere in some small disk around the North Pole.

$$\left\{ z \mid |z| > \frac{1}{\epsilon} \right\}$$

2.3 Continuity

We say that f is continuous at z_0 if $f(z_0)$ is defined, the limit $\lim_{z\to z_0} f(z)$ exists, and

$$\lim_{z\to z_0} f(z) = f(z_0)$$

$$\boxed{\begin{array}{c}\text{For every } \epsilon > 0 \text{, there exists } \delta > 0 \text{ such that for every } z \text{ in the domain of } f, \\ |z - z_0| < \delta \text{ implies } |f(z) - f(z_0)| < \epsilon\end{array}}$$

If f is continuous at all points in some set D, we say that f is continuous on/throughout D. For example, polynomials are continuous on/throughout $\mathbb{C}$.

Continuity theorems are the same from Real Analysis. If f and g are continuous at z_0, then

i) $f + g$ is continuous at z_0

ii) $f \cdot g$ is continuous at z_0

iii) f/g is continuous at z_0, assuming $g(z_0) \neq 0$

Theorem: Suppose g is continuous at z_0 and f is continuous at $g(z_0)$. Then $f \circ g$ is continuous at z_0.

Proof: We have to show that $\lim_{z \to z_0} f(g(z)) = f(g(z_0))$. Let $\epsilon > 0$. Since f is continuous at $g(z_0)$, there exists $\delta_1 > 0$ such that for all w in the domain of f,

$$|w - g(z_0)| < \delta_1 \implies |f(w) - f(g(z_0))| < \epsilon$$

Since g is continuous at z_0, there exists $\delta > 0$ such that for all z in the domain of g,

$$|z - z_0| < \delta \implies |g(z) - g(z_0)| < \delta_1$$

Thus for all $\epsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies |f(g(z)) - f(g(z_0))| < \epsilon$$

Theorem: Suppose $z_0, w_0 \in \mathbb{C}$ and f is continuous at z_0. If $f(z_0) \neq w_0$, then there exists a neighborhood N around z_0 such that $f(z) \neq w_0$ for all $z \in N$.

Essentially, if $f(z_0) \neq w_0$, we can find a neighborhood around z_0 such that f is not w_0 for any points in that neighborhood.

Proof: Since $f(z_0) \neq w_0$, there exists $\epsilon > 0$ such that

$$|w_0 - f(z_0)| > \epsilon$$

Since f is continuous at z_0, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies |f(z) - f(z_0)| < \epsilon$$

Let N be the δ-neighborhood around z_0. The above two statements imply that no $z \in N$ satisfies $f(z) = w_0$.

2.3.1 Uniform Continuity

If f is a complex function, continuous on a closed and bounded D, then $|f|$ attains a maximum on D.

If f is continuous on a closed and bounded set D, then $|f|$ is uniformly continuous on D. In epsilon-delta form,

$$\boxed{\begin{array}{c} f \text{ is uniformly continuous on } D \text{ if } \textit{for all } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that} \\ \text{for every } z_1, z_2 \in D,\ |z_1 - z_2| < \delta \text{ implies } |f(z_1) - f(z_2)| < \epsilon \end{array}}$$

The difference with normal continuity is that uniform continuity has a "universal delta" that applies to *all* ϵ, not just *each* ϵ.

Note: A closed and bounded set is called "compact."

3 Week 3 (Jan 26 – 30)

3.1 Derivatives

Let f be defined in some neighborhood of z_0. We define the derivative of f at z_0 similarly to the real-valued counterpart:

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \tag{option 1}$$

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \tag{option 2}$$

We can also denote the derivative w.r.t z as $\frac{d}{dz} f$.

The same properties hold as for real-valued derivatives:

i) $(f + g)'(z_0) = f'(z_0) + g'(z_0)$

ii) $(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$

iii) $(f/g)'(z_0) = \frac{f'(z_0)g(z_0) - f(z_0)g'(z_0)}{g(z_0)^2}, \quad g(z_0) \neq 0$

iv) $(cf)'(z_0) = cf'(z_0)$

v) $\frac{d}{dz} c = 0$

vi) $\frac{d}{dz} z^n = nz^{n-1}, \quad n \in \mathbb{Z}^+$

Theorem (Chain Rule): Suppose g is differentiable at z_0 and f is differentiable at $g(z_0)$. Then $f \circ g$ is differentiable at z_0, and

$$(f \circ g)'(z_0) = f'(g(z_0)) \cdot g'(z_0)$$

Proof: Using the definition of the derivative,

$$\begin{aligned} \lim_{h \to 0} \frac{f(g(z_0 + h)) - f(g(z_0))}{h} &= \lim_{h \to 0} \frac{f(g(z_0 + h)) - f(g(z_0))}{g(z_0 + h) - g(z_0)} \frac{g(z_0 + h) - g(z_0)}{h} \\ &= \lim_{h \to 0} \frac{f(g(z_0 + h)) - f(g(z_0))}{g(z_0 + h) - g(z_0)} \cdot g'(z_0) \end{aligned}$$

It suffices to show that

$$\lim_{h\to 0} \frac{f\left(g(z_0+h)\right) - f\left(g(z_0)\right)}{g(z_0+h) - g(z_0)} = f'\left(g(z_0)\right)$$

Let $\epsilon > 0$. Since f is differentiable $g(z_0)$, there exists $\delta_1 > 0$ such that for all w in the domain of f,

$$|w - g(z_0)| < \delta_1 \implies \left|\frac{f(w) - f(g(z_0))}{w - g(z_0)} - f'(g(z_0))\right| < \epsilon$$

We also know that g is continuous at z_0 (since it's differentiable there), i.e. $\lim_{h\to 0} g(z_0+h) = g(z_0)$. So there exists $\delta > 0$ such that

$$|h| < \delta \implies |g(z_0+h) - g(z_0)| < \delta_1$$

Let $w = g(z_0+h)$, a point in the domain of f. Then for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|h| < \delta \implies |g(z_0+h) - g(z_0)| \implies \left|\frac{f(w) - f(g(z_0))}{w - g(z_0)} - f'(g(z_0))\right| < \epsilon$$

So we've shown that

$$\lim_{h\to 0} \frac{f\left(g(z_0+h)\right) - f\left(g(z_0)\right)}{g(z_0+h) - g(z_0)} = f'\left(g(z_0)\right)$$
$$\lim_{h\to 0} \frac{f\left(g(z_0+h)\right) - f\left(g(z_0)\right)}{h} = f'(g(z_0)) \cdot g'(z_0)$$

Complex functions are often not differentiable. E.g. Let f be differentiable and nonzero at z_0. Then $g(z) = \overline{z} f(z)$ is not differentiable at z_0.

If we try to take the derivative, we get

$$\begin{aligned} g'(z_0) &= \lim_{h\to 0} \frac{(\overline{z_0+h}) f(z_0+h) - \overline{z_0} f(z_0)}{h} \\ &= z_0 \left(\frac{f(z_0+h) - f(z_0)}{h}\right) + \frac{\overline{h}}{h} f(z_0+h) \end{aligned}$$

The specific "path" we take to approach $h \to 0$ shouldn't matter if the derivative is well-defined. However,

$$\lim_{h\to 0} \frac{\overline{h}}{h} f(z_0+h) = f(z_0) \qquad (h \text{ is real})$$
$$\lim_{h\to 0} \frac{\overline{h}}{h} f(z_0+h) = -f(z_0) \qquad (h \text{ is imaginary})$$

Thus $g(z) = \overline{z} f(z)$ isn't differentiable at z_0.

3.2 Cauchy-Riemann Equations

The Cauchy-Riemann equations must be satisfied by any differentiable function. It's a necessary condition for the differentiability of f. Later, we'll make it a sufficient condition.

Theorem: Suppose $f(z) = u(x, y) + iv(x, y)$, $z_0 = x_0 + iy_0$, and f is differentiable at z_0. Then the function must satisfy

$$\begin{cases} u_x(x_0, y_0) = v_y(x_0, y_0) \\ u_y(x_0, y_0) = -v_x(x_0, y_0) \end{cases}$$

Proof: Since f is differentiable at z_0, the following limit must exist.

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

Let $h = s + it$. Then by substitution,

$$\begin{aligned} f(z_0 + h) &= f((x_0 + s) + i(y_0 + t)) \\ &= u(x_0 + s, y_0 + t) + iv(x_0 + s, y_0 + t) \end{aligned}$$

Now we try to find the limit:

$$\begin{aligned} f'(z_0) &= \lim_{s+it \to 0} \frac{f(z_0 + s + it) - f(z_0)}{s + it} \\ &= \lim_{s+it \to 0} \left(\frac{u(x_0 + s, y_0 + t) - u(x_0, y_0)}{s + it} + i \frac{v(x_0 + s, y_0 + t) - v(x_0, y_0)}{s + it} \right) \end{aligned}$$

We can restrict $s + it$ to the real axis, so $t = 0$. Then the derivative is

$$\begin{aligned} f'(z_0) &= \lim_{s \to 0} \left(\frac{u(x_0 + s, y_0) - u(x_0, y_0)}{s} \right) + i \lim_{s \to 0} \left(\frac{v(x_0 + s, y_0) - v(x_0, y_0)}{s} \right) \\ &= u_x(x_0, y_0) + iv_x(x_0, y_0) \end{aligned}$$

We can also restrict $s + it$ to the imaginary axis, so $s = 0$. Then the derivative is

$$\begin{aligned} f'(z_0) &= \lim_{t \to 0} \left(\frac{u(x_0, y_0 + t) - u(x_0, y_0)}{it} \right) + i \lim_{s \to 0} \left(\frac{v(x_0, y_0 + t) - v(x_0, y_0)}{it} \right) \\ &= -iu_y(x_0, y_0) + v_y(x_0, y_0) \end{aligned}$$

For the derivative to be well-defined, both results must be equal, thus

$$\begin{cases} u_x(x_0, y_0) = v_y(x_0, y_0) \\ u_y(x_0, y_0) = -v_x(x_0, y_0) \end{cases}$$

We can also make the Cauchy-Riemann equations a sufficient condition for differentiability by adding an extra condition.

Let $f(z) = u(x, y) + iv(x, y)$ be defined in a neighborhood of $z_0 = x_0 + iy_0$. Suppose u_x, u_y, v_x, and v_y are exist throughout the neighborhood and are continuous at (x_0, y_0).

Then f is differentiable at z_0 if and only if u and v satisfy the Cauchy-Riemann equations at (x_0, y_0) .

There's also a polar version of the Cauchy-Riemann equations, which can be derived with $x = r\cos\theta$ and $y = r\sin\theta$.

Theorem: If $f(z) = u(r,\theta) + iv(r,\theta)$ and f is differentiable at $z_0 = r_0 e^{i\theta_0}$, then

$$\begin{cases} u_r(r_0,\theta_0) = \frac{1}{r_0} v_\theta(r_0,\theta_0) \\ \frac{1}{r_0} u_\theta(r_0,\theta_0) = -v_r(r_0,\theta_0) \end{cases}$$

Moreover,

$$f'(z_0) = e^{-i\theta_0}\big(u_r(r_0,\theta_0) + iv_r(r_0,\theta_0)\big)$$

Proof: Rewrite the partial derivatives as

$$\begin{aligned} &1) \quad \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = u_x\cos\theta + u_y\sin\theta \\ &2) \quad \frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta} = -u_x r\sin\theta + u_y r\cos\theta \end{aligned}$$

Similarly for v,

$$\begin{cases} v_r = v_x\cos\theta + v_y\sin\theta \\ v_\theta = -v_x r\sin\theta + v_y r\cos\theta \end{cases}$$

If we use the Cauchy-Riemann conditions $u_x = v_y$ and $u_y = -v_x$, it becomes clear that $ru_r = v_\theta$ and $u_\theta = -rv_r$.

3.3 Analytic Functions

We say f is analytic on some set S if there exists an open set A such that $S \subseteq A$ and f is differentiable at each point of A.

In other words, f is analytic on S if every point in S has a neighborhood on which f is differentiable.

i) If S is open, then "f is analytic on S" just means "f is differentiable at each point of S."

ii) "f is analytic at z_0" just means "f is differentiable in a neighborhood of z_0."

Note: "Analytic at z_0" is a stronger condition than "differentiable at z_0." Analyticity tells us something about the function's behavior.

Properties of analytic functions:

i) If f and g are analytic at D, then so are $f+g$, $f\cdot g$, and f/g (as long as $g(z) \neq 0$ in D).

ii) If g is analytic on D and f is analytic on the image of g, then $f \circ g$ is analytic on D.

If f is analytic throughout $\mathbb{C}$, then it's called "entire." E.g. polynomials.

We call z_0 a singular point (singularity) of f is f is not analytic at z_0, but in every neighborhood of z_0 there exists a point at which f is analytic.

- For example, 0 is a singularity for $f(z) = 1/z$.

- However, 0 isn't a singularity of $g(z) = |z|^2$, because $g(z)$ isn't analytic anywhere.

Theorem: Suppose f is analytic on D and $f'(z) = 0$ for all of $z \in D$. Then f is constant on D.

Proof: From the Cauchy-Riemann equations,

$$f'(z) = \begin{cases} u_x(x,y) + iv_x(x,y) \\ v_y(x,y) - iu_y(x,y) \end{cases}$$

Since $f'(z) = 0$ on D, $u_x = u_y = v_x = v_y = 0$ on D. It follows that the gradient $\nabla u = \nabla v = \vec{0}$ on D.

Let $r(t) = (x(t), y(t))$ for $0 \leq t \leq 1$ be a parametrized line in D, from $z_0 = x_0 + iy_0$ to $z_1 = x_1 + iy_1$. Then $u(r(t))$ is a real-valued function with $u(r(0)) = u(x_0, y_0)$ and $u(r(1)) = u(x_1, y_1)$

$$\frac{du}{dt} = u_x \frac{dx}{dt} + u_y \frac{dy}{dt} = \nabla u \cdot r'(t) = 0$$

Thus $u(r(t))$ must be constant. Since the parametrized line was a general one on D, we conclude that u must be constant on D. Similar reasoning for v shows that $f = u + iv$ is constant on D.

3.4 Harmonic Functions

The Cauchy-Riemann equations are very restrictive conditions for a function to be analytic on a region. Which functions can be real or imaginary components of an analytic function?

Theorem: If $f = u + iv$ is analytic on D, and the 1st and 2nd partials of u and v are continuous throughout D, then

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ v_{xx} + v_{yy} = 0 \end{cases}$$

Proof: From Cauchy-Riemann, $u_x = v_y$ and $u_y = -v_x$ throughout D. Thus

$$(u_x)_y = (v_y)_x = (v_x)_y = (-u_y)_y$$

Similar reasoning for v can show that $v_{xx} = -v_{yy}$.

Functions that satisfy the above condition are called harmonic. More specifically, suppose ϕ is a real-valued function of 2 real variables x and y. Further, let ϕ have continuous 1st and 2nd partials onn D. Then ϕ is harmonic on D if

$$\phi_{xx} + \phi_{yy} = 0 \qquad \text{(Laplace's equation)}$$

Cauchy-Riemann equations imply that only harmonic functions can be real or imaginary parts of analytic functions.

$$\boxed{\text{If } f = u + iv \text{ is analytic on } D\text{, then } u \text{ and } v \text{ are harmonic on } D.}$$

If u, v are harmonic on D, and u, v satisfy the Cauchy-Riemann equations, then v is a *harmonic conjugate* of u.

Properties:

i) The harmonic conjugate isn't symmetric. "v is the harmonic conjugate of u" doesn't imply that "u is the harmonic conjugate of v."

ii) If v is the harmonic conjugate of u on D, then $-u$ is the harmonic conjugate of v on D.

iii) If v_1, v_2 are both harmonic conjugates of u on D, then $v_1 - v_2$ is constant throughout D.

iv) If u, v are both harmonic conjugates of each other on D, then u, v are both constant throughout D.

Theorem: $f = u + iv$ is analytic on D iff. v is a harmonic conjugate of u on D.

Theorem: If u is harmonic on a simply connected domain D, then there exists a harmonic conjugate v on D.

4 Week 4 (Feb 2 – 6)

4.1 Complex Expontential

We define the complex exponential function as

$$f(z) = e^x (\cos y + i \sin y), \quad z = x + iy$$

This is the counterpart of the real-valued exponential function $f(x) = e^x$. In fact, when $z = x \in \mathbb{R}$, the complex exponential reduces to e^x.

Properties of the complex exponential:

i) f is entire: it's analytic throughout $\mathbb{C}$. Thus Re(z) and Im(z) have continuous partials satisfying the Cauchy-Riemann equations.

ii) $f' = f$, to see this:
$$f'(z) = u_x + iv_x = e^x \cos y + ie^x \sin y = f(z)$$

iii) $|e^z| = e^x$, assuming $e^z \neq 0$.

iv) $\arg(z) = \{y + 2\pi n \mid n = 0, \pm 1, \pm 2, \dots\}$

v) $e^{z_1 + z_2} = e^{z_1} e^{z_2}$ for any $z_1, z_2 \in \mathbb{C}$

vi) $(e^x)^n = e^{nz}$ for any integer n.

Note that there are differences with the real-valued exponential. For example, $e^x > 0$ for $x \in \mathbb{R}$, but e^z can be negative for $z \in \mathbb{C}$. The classic example is $e^{\pi i} = -1$.

Additionally, e^z is *not* one-to-one. As seen in property (iv), it has period $2\pi i$. So no single-valued inverse exists, but we'll see a multi-valued $\log$ later.

There are potential ambiguities in notation. For example, given e^{iy}, should that be interpreted as a complex exponential? In this case both Euler's formula and the complex exponential return the same

answer:

$$e^{iy} = \cos y + i \sin y \tag{Euler}$$
$$e^{iy} = e^0 (\cos y + i \sin y) \tag{Exponential}$$

However, consider $e^{1/n}$. This could either mean the complex exponential, or the roots of e.

$$e^{1/n} = \{z \in \mathbb{C} \mid z^n = e\} \tag{Roots}$$
$$e^{1/n} = e^{1/n} (\cos 0 + i \sin 0) = \sqrt[n]{e} \tag{Exponential}$$

This semester, the convention will be that

If $z \neq e$, then $z^{1/n}$ denotes the n n-th roots of z.

If $z = e$, then $z^{1/n}$ denotes $\sqrt[n]{e}$.

4.2 Complex Trigonometric Functions

We also define the complex trig functions as follows:

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$
$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

These reduce to the real-valued $\sin$ and $\cos$ if $z = x \in \mathbb{R}$.

Properties (for all $z \in \mathbb{C}$):

i) $\sin(-z) = \sin(z)$
$\cos(-z) = \cos(z)$

ii) $\frac{d}{dz} \sin z = \cos z$
$\frac{d}{dz} \cos z = -\sin z$

iii) $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$
$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$

iv) $\sin^2 z + \cos^2 z = 1$

v) $\sin$ and $\cos$ are both periodic with period 2π.

There are some differences from the real-valued $\sin$ and $\cos$. For example, if $x \in \mathbb{R}$, then $|\sin x| \leq 1$ and $|\cos x| \leq 1$. However, in $\mathbb{C}$, the trig functions aren't bounded.

We can also write the complex trig functions in terms of the hyperbolic sine and cosine. Recall that

$$\sinh y = \frac{e^y - e^{-y}}{2} \quad , \quad \cosh y = \frac{e^y + e^{-y}}{2}$$

Thus for $z = x + iy$,

$$\sin(z) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$$
$$\cos(z) = \cos(x)\cosh(y) - i\sin(x)\sinh(y)$$

Note: the hyperbolic sine and cosine aren't equal to the complex sine and cosine. They look pretty similar, though.

4.3 Complex Logarithm

Given $z \neq 0$, we define the complex logarithm as

$$\log z = \{\ln|z| + i\theta \mid \theta \in \arg(z)\}$$

The complex logarithm is the inverse of the complex exponential. $w \in \log z$ iff. $e^w = z$.

The complex logarithm is a *set* of values, not a single value. This makes sense, as the complex exponential is multi-valued. So for some $z = re^{i\theta}$, we have

$$\begin{aligned}\log z &= \ln r + i\theta \\ &= \ln r + i(\theta + 2\pi) \\ &= \ln r + i(\theta + 4\pi) \\ &= \cdots\end{aligned}$$

So there are some subtleties in notation. For example, $z = e^{\log z}$ doesn't make sense because the log is multi-valued. Instead, we say $z \in e^{\log z}$. Additionally, the equation $\log e^z = \ln|e^z| + i\arg(e^z)$ is describing an equality of *sets* generated by $\log$ and $\arg$.

We can also define a single-valued logarithm. Recall that the principal argument $\text{Arg}(z)$ is the unique $\theta \in \arg(z)$ such that $-\pi < \theta < \pi$. Then the single-valued principal logarithm is

$$\text{Log}(z) = \ln|z| + i\text{Arg}(z)$$

Alternatively, we can define a branch of the multi-valued complex log to make it single-valued. Denote the branch as

$$L_\alpha(z) = \ln r + i\theta \quad , \quad r > 0, \alpha < \theta < \alpha + 2\pi$$

The branch is defined, single-valued, and analytic at all points in its domain excluding $\theta = \alpha$. The "normal" complex logarithm corresponds to $L_{-\pi}$.

$$L_{-\pi} = \ln r + i\theta \quad , \quad r > 0, -\pi < \theta \leq \pi$$

Theorem: The branch of the complex log

$$L_\alpha(z) = \ln r + i\theta \quad , \quad r > 0, \alpha < \theta < \alpha + 2\pi$$

is analytic at every point in its domain of definition, and its derivative at any such z is $1/z$.

Proof: Let $z = u(r,\theta) + iv(r,\theta)$. Then

$$\begin{aligned}\text{Re}\,(L_\alpha(z)) &= u(r,\theta) = \ln r \\ \text{Im}\,(L_\alpha(z)) &= v(r,\theta) = \theta\end{aligned}$$

u and v satisfy the polar version of Cauchy-Riemann, since

$$u_r = \frac{1}{r}v_\theta \quad , \quad \frac{1}{r}u_\theta = -v_r$$

Thus the branch L_α is analytic. Further, for every function $f(z) = u(r,\theta) + iv(r,\theta)$, we have

$$f'(z) = e^{-i\theta}\left(u_r(r,\theta) + iv_r(r,\theta)\right)$$

Substituting $f(z) = L_\alpha(z)$,

$$\frac{d}{dz}L_\alpha(z) = e^{-i\theta}\left(\frac{1}{r} + i\cdot 0\right) = \frac{1}{re^{i\theta}} = \frac{1}{z}$$

We've shown that L_α is analytic, and its derivative at any point z is $1/z$.

Properties of the complex log:

1. $\log(z_1 z_2) = \log(z_1) + \log(z_2)$ for all $z_1, z_2 \in \mathbb{C}$ and $z_1, z_2 \neq 0$
   - This is really a statement about sets.
   - This doesn't hold for log branches $L_\alpha(z)$.
2. $\log(z_1/z_2) = \log(z_1) - \log(z_2)$ for $z_1, z_2 \neq 0$
3. $\log(z^n) \neq n\log(z)$ in general for $z \neq 0$ and $n \in \mathbb{Z}$
   - A difference with the real logarithm.
   - However, $e^{n\log z} = e^{\log(z^n)} = z^n$. Equal when exponentiated.
4. $z^{1/n} = e^{\frac{1}{n}\log z}$ for any nonzero z and integer n
   - Recall that $z^{1/n} = \{w \mid w^n = z\}$, so this is another statement about sets.

5 Week 5 (Feb 9 – 13)

5.1 Generalized Exponentiation

Suppose $z, c \in \mathbb{C}$ and $z \neq 0$. Then we define the general complex exponent as

$$z^c = e^{c\log z}$$

Note that since the complex log is multi-valued, the exponent is also multi-valued.

For example, 1^i can be found as

$$\begin{aligned}1^i &= e^{i\log 1} \\ &= e^{i(\ln 1 + i(2\pi n))}\ ,\ n = 0, \pm 1, \pm 2, \ldots \\ &= e^{-2\pi n}\ ,\ n = 0, \pm 1, \pm 2, \ldots\end{aligned}$$

Setting $n = 0$ gives us the real-valued exponent.

To obtain a single-valued exponent, we use a branch of the complex log.

$$L_\alpha(z) = \ln(r) + i\theta \quad | \quad r > 0, \alpha < \theta < \alpha + 2\pi$$
$$z^c = e^{cL_\alpha(z)} \qquad \text{(branch of } z^c\text{)}$$

Theorem: The branch of $f(z) = z^c$ is analytic on its domain, and $f'(z) = cz^{c-1}$, where z^{c-1} is defined using the same branch L_α.

Proof: The domain of $f(z) = z^c = e^{cL_\alpha(z)}$ is the same as the domain of L_α. Since L_α is analytic on its domain and the exponential function is analytic everywhere, $f(z) = z^c = e^{cL_\alpha(z)}$ is analytic on its domain.

From the chain rule,

$$\begin{aligned}
\frac{d}{dz} z^c &= \frac{d}{dz} e^{cL_\alpha(z)} \\
&= e^{cL_\alpha(z)} \cdot \frac{c}{z} \\
&= c \cdot \frac{e^{cL_\alpha(z)}}{e^{L_\alpha(z)}} \\
&= ce^{(c-1)L_\alpha(z)} \\
&= cz^{c-1}
\end{aligned}$$

The branch here is defined using the same L_α branch as for $f(z) = z^c$.

5.2 Calculus of Complex Functions (of Real Variables)

Eventually, we want to compute integrals of the form

$$\int_C f(z)\, dz$$

where f is a complex-valued function and C is a curve in $\mathbb{C}$. Note that we can parametrize complex-valued functions in terms of real variables: e.g. $w(t) = u(t) + iv(t)$

5.2.1 Differentiation

If u, v are defined and differentiable on some $a \leq t \leq b$, then $w(t) = u(t) + iv(t)$ is differentiable on that interval, and the derivative is

$$w'(t) = u'(t) + iv'(t)$$

Suppose $z_0 \in \mathbb{C}$, and $w_1(t) = u_1(t) + iv_1(t)$ and $w_2(t) = u_2(t) + iv_2(t)$ are differentiable, complex-valued functions on $a \leq t \leq b$. Then for all $a \leq t \leq b$,

i) $\frac{d}{dt}(w_1 + w_2)(t) = w_1'(t) + w_2'(t)$

ii) $\frac{d}{dt}(z_0 w_1(t)) = z_0 w_1'(t)$

iii) $\frac{d}{dt}(w_1 \cdot w_2)(t) = w_1'(t)w_2(t) + w_1(t)w_2'(t)$

iv) $\frac{d}{dt}e^{z_0 t} = z_0 e^{z_0 t}$

Most of the same properties in the real-valued counterpart hold in the complex-valued realm as well. However, not all of them. For example, the mean-valued theorem doesn't hold for complex functions. You can see this with $w(t) = e^{it} = \cos t + i \sin t$, for $0 \leq t \leq 2\pi$. The derivative is $w'(t) = ie^{it}$.

$$|w'(t)| = 1 \quad , \quad \frac{w(2\pi) - w(0)}{2\pi - 0} = 0$$
$$w'(t) \neq \frac{w(2\pi) - w(0)}{2\pi - 0}$$

5.2.2 Integration

If $w(t) = u(t) + iv(t)$, then

$$\int_a^b w(t)\ dt = \int_a^b u(t)\ dt + i \int_a^b v(t)\ dt$$

$w(t)$ is integrable on $a \leq t \leq b$ if both u and v are integrable on $a \leq t \leq b$, i.e. if u and v are piecewise continuous on $a \leq t \leq b$.

Theorem (Fundamental Theorem of Calculus): Let $w_1(t) = u_1(t) + iv_1(t)$ and $w_2(t) = u_2(t) + iv_2(t)$. Suppose w_1, w_2 are continuous and $w_2'(t) = w_1(t)$ for all $a \leq t \leq b$. Then

$$\int_a^b w_1(t)\ dt = w_2(b) - w_2(a)$$

Proof: $w_2' = w_1$ implies $u_2' = u_1$ and $v_2' = v_1$. Therefore,

$$\begin{aligned}\int_a^b w_1(t)\ dt &= \int_a^b u_1(t)\ dt + i \int_a^b v_1(t)\ dt \\ &= u_2(b) - u_2(a) + i(v_2(b) - v_2(a)) \\ &= w_2(b) - w_2(a)\end{aligned}$$

This is the Fundamental Theorem of Calculus for complex-valued functions of a real variable.

5.2.3 Arcs

We call a curve $w(t) = u(t) + iv(t)$ with $a \leq t \leq b$ an *arc* if both u, v are continuous on $a \leq t \leq b$. An arc is notated $z(t) = x(t) + iy(t)$, for $a \leq t \leq b$.

- An arc is *simple* if it doesn't cross itself: $z(t_1) \neq z(t_2)$ unless $t_1 = t_2$.
- An arc is a *simple closed curve* (Jordan curve) if only its endpoints meet: $z(t_1) = z(t_2)$ iff. $\{t_1, t_2\} = \{a, b\}$.

Arcs have a natural orientation, given by the direction traversed by its output as the input increases.

In general, $\int_C f(z)\ dz$ is independent of the arc C's parametrization. However, the signs flip if the orientations are different:

$$\int_C f(z)\ dz = -\int_{-C} f(z)\ dz$$

An arc is differentiable if $x(t), y(t)$ have continuous derivatives for all $a \leq t \leq b$. If $z(t) = x(t) + iy(t)$ is a differentiable arc, its length is defined as

$$L = \int_a^b |z'(t)|\, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2}\, dt$$

A differentiable arc is *smooth* if $z'(t)$ is continuous on $a \leq t \leq b$ and nonzero for $a < t < b$.

A *contour* is an arc composed of a finite number of smooth arcs, joined end-to-end. Thus a contour is like a "piecewise smooth" arc.

A *simple closed contour* $z(t)$ for $a \leq t \leq b$ is a contour that's also a simple closed curve. That is, $z(a) = z(b)$, and there's no other intersections.

Theorem (Jordan Curve Theorem): A set of points on a simple closed contour C is the boundary of 2 disjoint domains—one bounded, one not.

It helps to think of the bounded points as the "interior" of the shape defined by the contour. The unbounded points are the "exterior."

Proof: Too hard.

6 Midterm Week (Feb 16 – 20)

7 Week 7 (Feb 23 – 27)

7.1 Antiderivatives

Let f be a continuous function on the domain D. A function F satisfying $F'(z) = f(z)$, for all $z \in D$, is called an antiderivative of f on D. Note that the antiderivative is necessarily analytic, since $f(z)$ is continuous.

Theorem: If F and G are both antiderivatives of f on D, then for some constant c, $F(z) = G(z) + c$ for all $z \in D$. That is, antiderivatives differ only by a constant.

Proof: Let $H(z) = F(z) - G(z)$. The difference of two analytic functions is analytic, thus $H(z)$ is analytic on D. Then $H'(z) = F'(z) - G'(z) = f(z) - f(z) = 0$ for all $z \in D$. Since $H'(z)$ is 0, $H(z) = c$ for some $c \in \mathbb{C}$.

7.2 Independence of Path in Contour Integration

There's a neat connection between the existence of an antiderivative and a contour integral that's independent of path.

Theorem: Suppose $f(z)$ is continuous in a domain D. If any one of the following is true, then so are the others.

a) $f(z)$ has an antiderivative $F(z)$ throughout D.

b) The integrals of $f(z)$ along contours lying entirely in D and extending from any fixed point z_1 to z_2 all have the same value given below, where $F(z)$ is the antiderivative.

$$\int_{z_1}^{z_2} f(z)\ dz = F(z)|_{z_1}^{z_2} = F(z_2) - F(z_1)$$

c) The integrals of $f(z)$ around closed contours lying entirely in D are all 0.

Proof (a implies b): Let F be the antiderivative of a continuous function f on D. Let C be some contour in D with endpoints α, β. Break this up into n smooth arcs $C_1, \ldots, C_n$.

$$\int_C f(z)\ dz = \sum_{k=1}^{n} \int_{C_k} f(z)\ dz$$

Suppose $z_k(t)$ is a smooth parametrization of C_k, where $a_k \leq t \leq b_k$. Then the contour integral of each smooth arc is

$$\int_{A_k} f(z)\ dz = \int_{a_k}^{b_k} f(z_k(t))z_k'(t)\ dt$$

Also note that $\frac{d}{dt}F(z_k(t)) = F'(z_k(t))z_k'(t) = f(z_k(t))z_k'(t)$. That is, $F(z_k(t))$ is the antiderivative of $f(z_k(t))z_k'(t)$. Thus, the integral above becomes

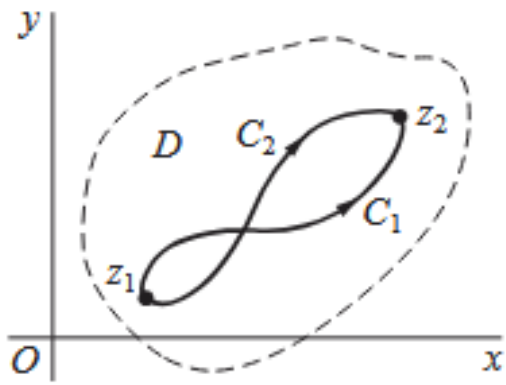
$$\int_{A_k} f(z)\ dz - F(z_k(b_k)) - F(z_k(a_k)) = F(z_k) - F(z_{k-1})$$

The sum of each contour integral collapses:

$$\int_C f(z)\ dz = \sum_{k=1}^{n} (F(z_k) - F(z_{k-1})) = F(z_n) - F(z_0) = F(\beta) - F(\alpha)$$

Thus statement (a) implies (b). The contour integral only depends on the endpoints, not the path C.

Proof (b implies c):



C_1 and C_2 as two contours on D (Brown & Churchill 145)

By (b), we know that

$$\int_{C_1} f(z)\, dz = \int_{C_2} f(z)\, dz$$

Now define the general closed contour C as $C = C_1 - C_2$. then

$$\int_C f(z)\, dz = \int_{C_1} f(z) \; - \int_{C_2} f(z)\, dz = 0$$

Proof (c implies a): First, note that (c) implies (b), since for every closed contour in D, we can split it into two distinct contours. Then all we have to do is show that (b) implies (a). From (b), we know that integration is independent of path in D, so the function

$$F(z) = \int_{z_0}^{z} f(s)\, ds$$

can be defined. We want to show that $F'(z) = f(z)$ for all $z \in D$. Let $z + \Delta z$ be a point also in D. Then

$$F(z + \Delta z) - F(z) = \int_{z_0}^{z+\Delta z} f(s)\, ds - \int_{z_0}^{z} f(s)\, ds = \int_{z}^{z+\Delta z} f(s)\, ds$$

Also note that

$$\int_{z}^{z+\Delta z} ds = \Delta z \quad \text{and} \quad f(z) = \frac{1}{\Delta z} \int_{z}^{z+\Delta z} f(z)\, ds$$

Thus we can write

$$\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z} \int_{z}^{z+\Delta z} [f(s) - f(z)]\; ds$$

Note that since f is continuous, for each $\epsilon > 0$, there exists $\delta > 0$ such that $|s - z| < \delta$ implies $|f(s) - f(z)| < \epsilon$. Thus, if $z + \Delta z$ is close enough to z such that $|\Delta z| < \delta$,

$$\left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| < \frac{1}{|\Delta z|} \cdot \epsilon |\Delta z| = \epsilon$$

We've shown that

$$\lim_{\Delta z \to 0} \frac{F(z + \Delta z) - F(z)}{\Delta z} = f(z)$$

so $F'(z) = f(z)$ throughout D.

7.3 Cauchy-Goursat Theorem

Previously, we saw that the contour integral of a continuous function $f(z)$ was 0 if $f(z)$ **has an antiderivative**. There are other conditions where the contour integral is 0. First, we give Cauchy's result. Next, we give the strengthened result proved by Goursat.

7.3.1 Cauchy's result

Theorem: If f is analytic at all points interior to and on the simple closed contour C, and f' is continuous there, then

$$\int_C f(z)\ dz = 0$$

Proof: Suppose $z(t)$ with $a \leq t \leq b$ is a parametrization of C.

$$\int_C f(z)\ dz = \int_a^b f(z(t))z'(t)\ dt$$

Now let $f(z) = u(x, y) + iv(x, y)$ and $z(t) = x(t) + iy(t)$. If we plug these into our integrand, we get

$$\begin{aligned}
\int_C f(z)\ dz &= \int_a^b \left[u(x(t), y(t)) + iv(x(t), y(t))\right] \left[x'(t) + iy'(t)\right]\ dt \\
&= \int_a^b (ux' - vy')\ dt + i \int_a^b (vx' + uy')\ dt \\
&= \int_C u\ dx - v\ dy + i \int_C v\ dx + u\ dy
\end{aligned}$$

Now recall Green's theorem from Multivariable Calculus, which holds if P, Q, and **their first-order partials are continuous throughout** R.

$$\oint_C P\ dx + Q\ dy = \iint_R (Q_x - P_y)\ dA$$

We can apply this to our integral since C is a closed contour. Let R be the region closed by C. Then

$$\int_C f(z)\ dz = \iint_R (-v_x - u_y)\ dA + i \iint_R (u_x - v_y)\ dA$$

In light of the Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$, the integral is 0. Thus, if $f(z)$ is analytic (satisfying Cauchy-Riemann) and f' is continuous (so that the partials are continuous for Green's theorem), then the contour integral is 0.

7.3.2 Cauchy-Goursat Theorem

Goursat proved that the condition of continuity on f' can be omitted. This can be used to show (for example) that the derivative f' of an analytic function f is also analytic without having the assume the continuity of f'.

Theorem: If $f(z)$ is analytic at all points interior to and on a simple closed contour C, then

$$\int_C f(z)\, dz = 0$$

Proof: Too technical and I'm lazy to write it out. Given in Brown & Churchill pp. 150 – 154.

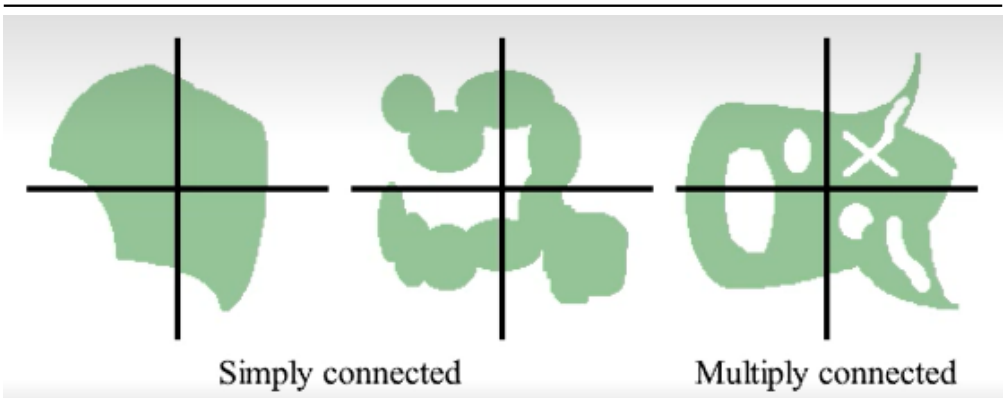
As an example, consider $f(z) = 1/z$, which is analytic for all $z \neq 0$. The contour integral of a unit circle, however, is

$$\int_C \frac{1}{z}\, dz = 2\pi i \neq 0$$

because there's a singularity at $z = 0$, which is a region within the circle.

7.4 Generalization of Cauchy-Goursat: Simply Connected Domains

We call a domain D **simply connected** if given any simple closed contour C in D, every point interior to C is also in D. Otherwise, we call D **multiply connected**.



Simply connected vs. multiply connected

Theorem: Let f be analytic on a simply connected domain D. If C is any closed contour lying in D, then

$$\int_C f(z)\, dz = 0$$

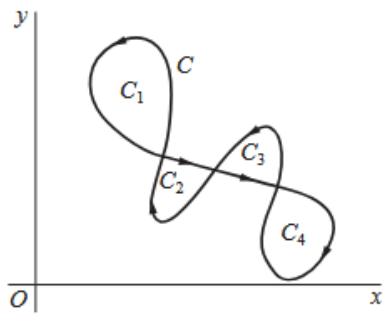
Proof: First, suppose C is a **simple** closed contour (doesn't intersect itself). Since D is simply connected, all points interior to C also lie in D. Then by Cauchy-Goursat for **simple** closed contours,

$$\int_C f(z)\, dz = 0$$

Now suppose that C is a **general** closed contour that might intersect itself a **finite** number of times. We can still write this as a sum of simple closed contours, e.g. $C = C_1 + C_2 + C_3 + C_4$. Then by Cauchy-Goursat for simple closed contours,

$$\int_C f(z)\, dz = \sum_k \int_{C_k} f(z)\, dz = 0$$

We won't cover the case of infinite self-intersections.



Contour that intersects itself a finite number of times

Theorem: If f is analytic on a simply connected domain D, then f has an antiderivative on D.

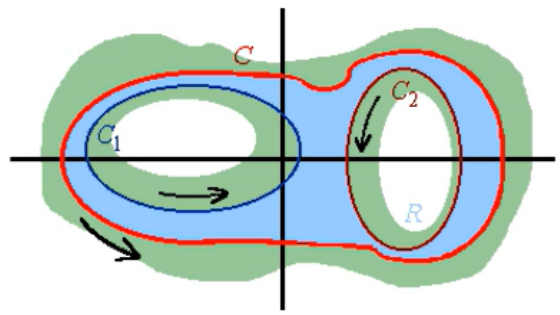
Proof: If f is analytic on D, then f is continuous on D. By Cauchy-Goursat, $\int_C f(z)\, dz = 0$ for any closed contour C in D. By the theorems in the "Independence of Path" section, f must have an antiderivative throughout D.

Note that the complex plane itself is simply connected, so if f is analytic throughout $\mathbb{C}$ ("entire"), then $\int_C f(z)\, dz = 0$, so f has an antiderivative everywhere.

7.5 Application of Cauchy-Goursat: Principle of Deformation of Paths

Let C be a simple closed contour oriented in the counterclockwise direction. Suppose C_1 and C_2 are simple closed contours oriented counterclockwise, which are disjoint, have disjoint interiors, and both lie interior to C. Suppose f is analytic on the closed region R consisting of all points on the interior of C except for points interior to C_1 and C_2. Then

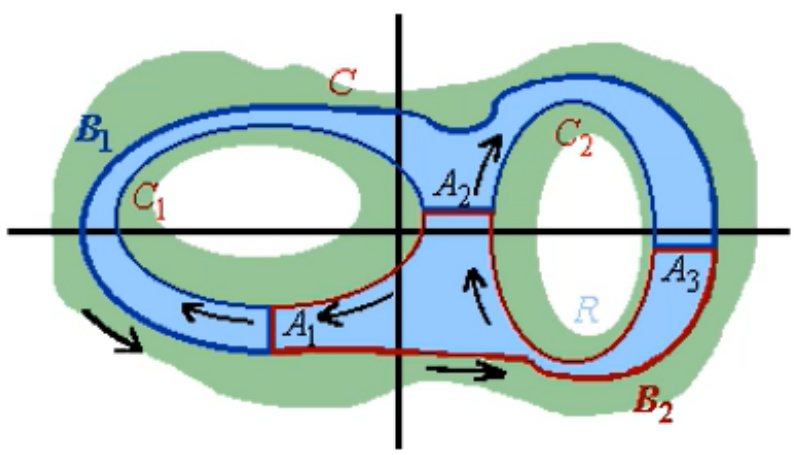
$$\int_C f(z)\, dz = \int_{C_1} f(z)\, dz + \int_{C_2} f(z)\, dz$$



Domain with holes

So we can find the contour integral around a domain with holes by summing the contour integrals around each hole.

Proof: Draw paths A_1, A_2, A_3 from C to C_1, C_1 to C_2, and C_2 to C. Let B_1, B_2 be the two simple closed contours which are the boundaries of the two parts of the region.



Setup of proof

$$\int_{B_1+B_2} f(z)\,dz = \int_C f(z)\,dz - \int_{C_1} f(z)\,dz - \int_{C_2} f(z)\,dz$$

Cauchy-Goursat implies that $\int_{B_1} f(z)\,dz = 0$ and $\int_{B_2} f(z)\,dz = 0$, thus

$$0 = \int_C f(z)\,dz - \int_{C_1} f(z)\,dz - \int_{C_2} f(z)\,dz$$
$$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz$$

This can also be proven for a general number of holes. It helps us find the contour integral of a domain on which f isn't analytic, but we know exactly where the "holes" are.

Corollary: Suppose C and C_1 are simple closed contours oriented counterclockwise, such that C_1 lies interior to C and f is analytic on C. Let f be analytic at all points which are both interior to C and exterior to C_1. Then

$$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz$$

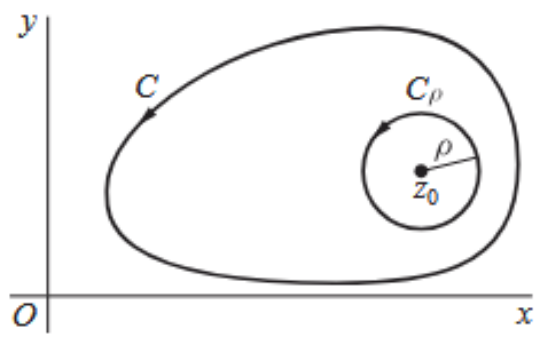
This is called the **principle of deformation of paths**. The value of the contour integral doesn't change if the closed contours can be continuously deformed into each other through points on which f is analytic.

7.6 Cauchy Integral Formula

Theorem: Let C be a simple closed contour oriented counterclockwise, and suppose f is analytic on and inside C. If z_0 is a point interior to C, then

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\,dz$$

Proof: Let C_p denote a circle $|z - z_0| < \rho$ oriented counterclockwise. ρ is small, so C_p is inside C.



Proof setup

The quotient $f(z)/(z-z_0)$ is analytic between and on the contours C_p and C. Thus, from the principle of deformation of paths,

$$\int_C \frac{f(z)}{z - z_0}\,dz = \int_{C_p} \frac{f(z)}{z - z_0}\,dz$$

So we can write

$$\int_C \frac{f(z)}{z - z_0}\,dz - f(z_0)\int_{C_p} \frac{1}{z - z_0}\,dz = \int_{C_p} \frac{f(z) - f(z_0)}{z - z_0}\,dz$$
$$\int_C \frac{f(z)}{z - z_0}\,dz - 2\pi i f(z_0) = \int_{C_p} \frac{f(z) - f(z_0)}{z - z_0}\,dz$$

Note that f is analytic and therfore continuous. At every point z_0, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|z - z_0| < \delta$ implies $|f(z) - f(z_0)| < \epsilon$. Let ρ be the radius of the circle C_p, and set it to

smaller than the current δ. Then $|z - z_0| = \rho < \delta$. Then we can bound the above RHS integral with a modulus:

$$\left| \int_{C_p} \frac{f(z) - f(z_0)}{z - z_0} \, dz \right| < \frac{\epsilon}{\rho} 2\pi\rho = 2\pi\epsilon$$

Since ϵ is arbitrary, it follows that

$$\int_C \frac{f(z)}{z - z_0} \, dz - 2\pi i f(z_0) = 0$$
$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} \, dz$$

The Cauchy Integral Formula implies that the value of f at each point interior to C is determined by values of f on C.

Corollary: Let C be a simple closed contour. Suppose f and g are analytic both on and inside C. If $f(z) = g(z)$ on C, then $f(z) = g(z)$ inside C.

7.7 Generalized Cauchy Integral Formula

Theorem: Suppose C is a simple closed contour oriented counterclockwise, and f is analytic at all points on and interior to C. If z is interior to C, then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(s)}{(s - z)^{n+1}} \, ds$$

Proof: Lengthy. Check book or Lecture 26 if interested (uses induction).

Corollary 1: If f is analytic at a point z, then so is $f^{(n)}$ for all $n = 1, 2, \ldots$

Proof: Suppose f is analytic at z, and let n be a positive integer. There exists a neighborhood of z on which f is analytic. Let C be a circle centered at z and lying within that neighborhood. By the theorem, $f^{(n+1)}$ exists at all points within C, thus $f^{(n)}$ is analytic at z.

Note that this is different from real functions: E.g. the existence of $f'(x)$ for a real-valued f doesn't guarantee the existence of $f''(x)$.

Corollary 2: If $f(z) = u(x, y) + iv(x, y)$ is analytic at $z = x + iy$, then all the partials of u and v are continuous at (x, y).

Proof: For every n, $f^{(n)}$ is analytic at z by Corollary 1. This implies that $f^{(n)}$ is continuous at z. Consider the first derivative:

$$f'(z) = u_x(x, y) + iv_x(x, y) = v_y(x, y) - iu_y(x, y)$$

Thus u_x, u_y, v_x, and v_y must all be continuous. Repeat the argument with f'', then f''', etc.

Corollary 3 (Morera's Theorem): Suppose f is continuous throughout a domain D and for every closed contour C lying in D,

$$\int_C f(z)\ dz = 0$$

Then f is analytic in D.

Proof: An earlier result implies that since the contour integral is 0, the antiderivative must exist and be analytic on D (since f is continuous).

$$F'(z) = f(z)\ ,\ \ z \in D$$

Corollary 1 implies that F' is analytic in D, so f must also be analytic in D.

Corollary 4: Suppose f is continuous on the simply connected domain D. Then f is analytic on D iff. for all closed contours C lying in D,

$$\int_C f(z)\ dz = 0$$

Return to the generalized Cauchy integral formula:

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(s)}{(s-z)^{n+1}}\ ds$$

Note that

$$\frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}\ dz = \frac{f^{(n)}(z_0)}{n!}$$

This looks familiar to a Taylor series term: specifically, the n-th coefficient of the Taylor series of f centered at z_0.

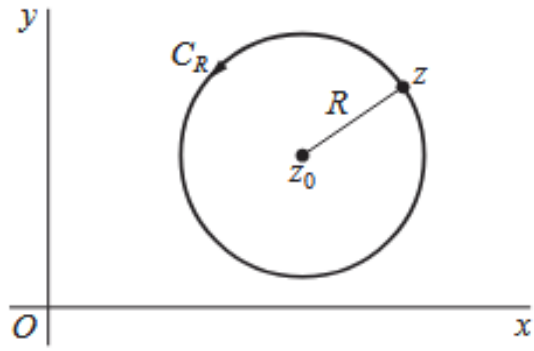
8 Week 8 (Mar 2 – 6)

8.1 Liouville's Theorem

8.1.1 Cauchy's Inequality

Theorem (Cauchy's Inequality): Let $z_0 \in \mathbb{C}$ and C_R be the circle $|z - z_0| = R$ oriented counter-clockwise. Suppose f is analytic on and inside C_R and that M_R is the maximum value of $|f|$ on C_R. Then

$$|f^{(n)}(z_0)| \leq \frac{n!M_R}{R^n}\ \ ,\ \ n = 1, 2, \ldots$$



Setup for Cauchy's Inequality (Brown & Churchill 170)

Proof: By Cauchy's Integral Formula,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_R} \frac{f(z)}{(z - z_0)^{n+1}} \, dz$$

We know that

$$\left| \frac{f(z)}{(z - z_0)^{n+1}} \right| < \frac{M_R}{R^{n+1}}$$

Thus

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi} \cdot \frac{M_R}{R^{n+1}} \cdot 2\pi R = \frac{n! M_R}{R^n} \quad , \quad n = 1, 2, \ldots$$

8.1.2 Liouville's Theorem

We can use Cauchy's inequality to prove Liouville's theorem.

Theorem (Liouville's Theorem): If f is entire and bounded on the whole complex plane, then f is constant.

Note that this isn't true for real-valued functions. E.g. $f(x) = \sin(x)$ is differentiable everywhere and bounded, but not constant.

Proof: Suppose $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Let $z_0 \in \mathbb{C}$ and C_R be the circle $|z - z_0| = R$. Let M_R be the maximum value of $|f|$ on C_R. Note that $M_R \leq M$.

Using Cauchy's inequality for $n = 1$,

$$|f'(z_0)| \leq \frac{M_R}{R} \leq \frac{M}{R}$$

Since R can be arbitrarily large, $|f'(z_0)| = 0$. Hence $f'(z_0) = 0$,and since z_0 was arbitrary, $f'(z) = 0$ for all $z \in \mathbb{C}$.

8.2 The Fundamental Theorem of Algebra

Theorem: If $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where $a_n \neq 0$, is a non-constant polynomial, then it has a zero in $\mathbb{C}$.

In other words, $\mathbb{C}$ is "algebraically closed": you don't need to go outside $\mathbb{C}$ to find roots of a complex polynomial. This doesn't hold for real-valued functions, e.g. $P(x) = x^2 + 1$ has no real roots.

Thus, there exists $a \in C$ and $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ such that

$$P(z) = a(z - \alpha_1) \ldots (z - \alpha_n)$$

Lemma: For some $R > 0$, the reciprocal $1/P(z)$ satisfies the inequality

$$\left| \frac{1}{P(z)} \right| < \frac{2}{|a_n| R^n} \quad \textit{if} \quad |z| > R$$

The proof is in p. 12 of the book.

Proof of Theorem: We'll use proof by contradiction. Suppose $P(z) \neq 0$ for any value of z. By the Lemma above, $1/P(z)$ is bounded in the region exterior to the disk $|z| \leq R$. But $1/P(z)$ is continuous on that closed disk, thus $1/P(z)$ is bounded there too. Hence $1/P(z)$ is bounded on the entire plane.

By Liouville's theorem, we expect that $1/P(z)$ and consequently $P(z)$ are constant. But $P(z)$ isn't constant. This is a contradiction.

8.3 Maximum Modulus Principle

Recall that a real-valued function which is continuous on a closed & bounded region must have a maximum there. A similar result holds for complex functions: if R is a closed and bounded region in $\mathbb{C}$ and f is continuous on R, then for some $z_0 \in R$, $|f(z_0)| \geq |f(z)|$ for all $z \in R$.

Now we'll prove that a non-constant function, when analytic on a closed & bounded region R, only takes it maximum value at boundary points of R. The maximum points are never within R.

First, a lemma is needed.

Lemma: Suppose $|f(z)| \leq |f(z_0)|$ at each point z in some neighborhood $|z - z_0| < \epsilon$, in which f is analytic. Then $f(z) = f(z_0)$ for all z in the neighborhood. I.e. f is constant there.

Proof: Assume f satisfies the conditions. Let z_1 be any point other than z_0 in the neighborhood. Let ρ be the distance between z_0 and z_1. Let C_ρ denote the circle $|z - z_0| = \rho$ oriented counterclockwise and centered at z_0. By the Cauchy Integral Formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{C_\rho} \frac{f(z)}{z - z_0} \, dz$$

If we let $z = z_0 + \rho e^{i\theta}$ for $0 \leq \theta \leq 2\pi$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta})\, d\theta$$
$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})|\, d\theta$$

But since $|f(z_0 + \rho e^{i\theta})| \leq |f(z_0)|$, we have

$$\int_0^{2\pi} |f(z_0 + \rho e^{i\theta})|\, d\theta \leq \int_0^{2\pi} |f(z_0)|\, d\theta = 2\pi |f(z_0)|$$
$$|f(z_0)| \geq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})|\, d\theta$$

Thus

$$|f(z_0)| = \frac{1}{2\pi} \int_0^{2\pi} |f(z_0) + \rho e^{i\theta}|\, d\theta$$
$$\int_0^{2\pi} |f(z_0)| - |f(z_0 + \rho e^{i\theta})|\, d\theta = 0$$

Since $|f(z_0 + \rho e^{i\theta})| \leq |f(z_0)|$, the integrand is always greater than or equal to 0. But the overall integral is 0, thus the integrand must always be 0. Hence, $|f(z_0 + \rho e^{i\theta})| = |f(z_0)|$, and $|f(z)| = |f(z_0)|$ for all points z on the circle $|z - z_0| = \rho$.

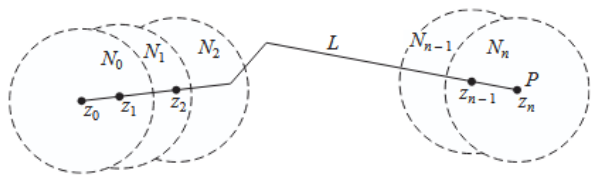
Now we prove the maximum modulus principle.

Theorem (Maximum Modulus Principle): If f is analytic and not constant in a given domain D, then $|f(z)|$ has no maximum value in D. That is, there's no point z_0 *within* D such that $|f(z)| \leq |f(z_0)|$ for all points $z \in D$.

Proof: We'll use proof by contradiction. Assume that f has a maximum value in D at z_0. Consider a polygonal line in D extending from z_0 to any other point P within D. Let d be the smallest distance from any point on L to the boundary of D. Let $z_0, \ldots, z_n$ be points lying along L such that z_n coincides with P and

$$|z_k - z_{k-1}| < d$$

This guarantees that the neighborhoods of each z_k lie within D as well. We've formed a finite sequence of neighborhoods $N_0, \ldots, N_n$ such that each N_k has center z_k and radius d. The center of each N_k lies in N_{k-1}.



Proof setup (Brown & Churchill 175)

Since $|f(z)|$ was assumed to have a maximum value at z_0, it also has a maximum value in N_0 at z_0. By the previous Lemma, f has the constant value $f(z_0)$ in N_0, thus $f(z_1) = f(z_0)$. Continue that until $f(z) = f(z_0)$.

Thus, $f(z)$ isn't a maximum. This is a contradiction, so there's no point *within* D that's a maximum. But if f is analytic at each point in D, then there must be some M such that $|f(z)| \leq M$ for all $z \in D$. The maximum isn't found on the interior. So it must be on the boundary.

Corollary: Suppose that f is continuous on a closed & bounded region R and that it's analytic and not constant in the interior of R. Then the maximum value of $|f(z)|$ in R occurs somewhere on the boundary of R, never the interior.

8.4 Basics of Sequences and Series

8.4.1 Sequences

A sequence of complex numbers $z_1, z_2, \ldots, z_n$ **converges** to z if

$$\boxed{\text{For all } \epsilon > 0 \text{ there exists an integer } k \text{ such that } |z - z_n| < \epsilon \text{ whenever } n > k.}$$

If $z_1, z_2, \ldots, z_n$ converges to z, we say that it's **limit** is z and write

$$\lim_{n\to\infty} z_n = z$$

Else, the sequence **diverges**.

Theorem: Suppose that for each positive integer n, $z_n = x_n + iy_n$ and $z = x + iy$. Then $\lim z_n = z$ if and only if $\lim x_n = x$ and $\lim y_n = y$.

Proof: Suppose $\lim z_n = z$ and $\epsilon > 0$. There exists k such that for $n > k$,

$$|(x_n + iy_n) - (x + iy)| = |z_n - z| < \epsilon$$
$$|(x_n - x) + i(y_n - y)| < \epsilon$$

Further, for all n, $|x_n - x| \leq |(x_n - x) + i(y_n - y)|$. (Remember: this is a modulus, not an absolute value). Thus for $n > k$,

$$|x_n - x| < \epsilon \ , \ |y_n - y| < \epsilon$$

For the other direction, suppose $\lim x_n = x$ and $\lim y_n = y$. We can find k_1, k_2 such that

$$\begin{aligned} |x_n - x| &< \epsilon/2 \ \textit{for} \ \ n > k_1 \\ |y_n - y| &< \epsilon/2 \ \textit{for} \ \ n > k_2 \end{aligned}$$

Let $k = k_1 + k_2$. Then for any $n > k$,

$$\begin{aligned} |z_n - z| = |x_n + iy_n - (x + iy)| &= |(x_n - x) + i(y_n - y)| \\ &\leq |x_n - x| + |y_n - y| \\ &< \epsilon/2 + \epsilon/2 = \epsilon \end{aligned}$$

Properties: If the sequence $z_1, z_n, \ldots, z_n$ converges to z and the sequence $w_1, w_2, \ldots, w_n$ converges to w, and c is any complex number, then

i) $\lim(z_n + w_n) = z + w$

ii) $\lim(z_n - w_n) = z - w$

iii) $\lim cz_n = cz$

8.4.2 Series

If $\sum z_n = z_1 + z_2 + \cdots$ is an infinite series, the k-th partial sum is given by

$$S_k = \sum_{n=1}^{k} z_n = z_1 + z_2 + \cdots + z_k$$

A series $\sum z_n$ is said to **converge** to the number S if

$$\lim_{k\to\infty} S_k = S$$

otherwise, the sum **diverges**.

The **tail** or **remainder** of the series is usually denoted ρ_k and given by $S - S_k = z_{k+1} + z_{k+2} + \cdots$

Theorem: If the series $\sum z_n$ converges, then $\lim z_n = 0$.

Proof: Let S denote the sum of the series, and S_k its k-th partial sum. Then

$$\begin{aligned} z_k &= S_k - S_{k-1} \\ \lim_{k\to\infty} z_k &= \lim_{k\to\infty} S_k - S_{k-1} = S - S = 0 \end{aligned}$$

Theorem: If $\sum z_n$ converges, the set $\{z_n \mid n = 1, 2, \ldots\}$ is bounded.

Proof: Set $\epsilon = 1$. We can find k such that $|z_n| < 1$ for all $n > k$. Let $M = 1 + \max\{|z_1|, |z_2|, \ldots, |z_k|\}$. Then for all n, $|z_n| < M$.

A series $\sum z_n$ is said to **converge absolutely** if $\sum |z_n|$ converges. If a series converges absolutely, then it also converges.

Let $z_0, a_0, a_1, a_2, \dots \in \mathbb{C}$. A **power series** centered at z_0 is given by

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

*Later, we'll build up to showing that an analytic function has a convergent complex power series and vice versa.

8.5 Taylor Series Representation of Analytic Functions

Theorem: Suppose f is analytic in the disk $|z - z_0| < R_0$. Then for each z in the disk,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad , \quad a_n = \frac{f^{(n)}(z_0)}{n!}$$

Note that this relies on an earlier theorem: if f is analytic at z_0, then $f^{(n)}(z_0)$ exists for $n = 1, 2, \dots$

If f is entire, the Taylor series representation holds throughout $\mathbb{C}$.

Proof: Quite long. See section 63 of Brown & Churchill.

Example representations:

$$\sin(z) = \sum \frac{(-1)^n z^{2n+1}}{(2n+1)!} \qquad \cos(z) = \sum \frac{(-1)^n z^{2n}}{(2n)!}$$

$$e^z = \sum \frac{z^n}{n!} \qquad \frac{1}{1-z} = \sum z^n \quad \text{(for } |z| < 1\text{)}$$

So these just match the Taylor series of their real-valued counterparts.

Finding the Taylor/Maclaurin series for a function is simple enough, just keep finding derivatives. But remember to check whether f is analytic in the region.

9 Week 9 (Mar 9 – 13)

9.1 Laurent Series

Recall that if f is analytic at z_0, then for all z in some neighborhood of z_0,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

If we allow negative powers of $(z - z_0)$, we can obtain a series representation of f even when f is not analytic at z_0. This general series is called a **Laurent series**.

Theorem (Laurent's theorem): Let C_1 and C_2 be the counterclockwise oriented circles $|z - z_0| = r_1$ and $|z - z_0| = r_2$, where $r_1 < r_2$. Suppose f is analytic in the region $r_1 \leq |z - z_0| \leq r_2$. Then in that region,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}$$

Where a_n and b_n are given by

$$a_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(s)}{(s - z_0)^{n+1}}\, ds \quad , \quad b_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(s)}{(s - z_0)^{-n+1}}\, ds$$

This is called a Laurent series of f centered at z_0.

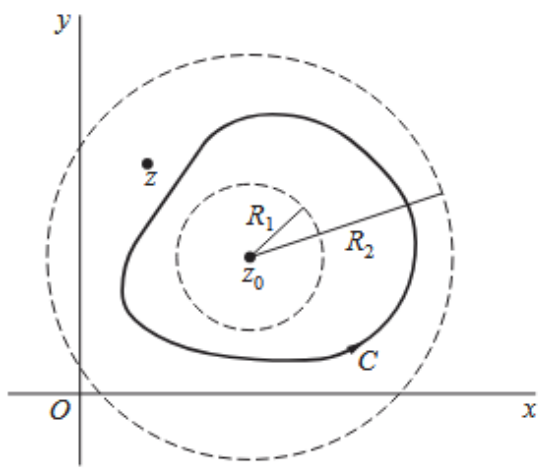


Diagram (Brown & Churchill 179)

Proof: See section 67 of the textbook. Uses Cauchy-Goursat and divides the cases of $z_0 = 0$ and $z_0 \neq 0$.

Theorem (more general): Suppose f is analytic in the annular domain $R_1 < |z - z_0| < R_2$, and let C be a counterclockwise oriented simple closed contour which goes around z_0 and lies in the domain. Then for every z in that domain,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}$$

Where a_n and b_n are given by

$$a_n = \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{n+1}}\, ds \quad , \quad b_n = \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{-n+1}}\, ds$$

The generalization is that f doesn't have to be analytic on $|z - z_0| = R_1$ and $|z - z_0| = R_2$. Furthermore, a_n and b_n are just integrals along the general curve C, not C_1 and C_2.

Proof: For any point z in the annular domain, draw circles C_1 and C_2 centered at z_0, such that z is between them. By Laurent's theorem above,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}$$
$$a_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(s)}{(s - z_0)^{n+1}}\, ds \quad , \quad b_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(s)}{(s - z_0)^{-n+1}}\, ds$$

The integrands of a_n and b_n are analytic throughout the annular domain, since $s \neq z_0$. By the principal of deformation of paths, they're equivalent to

$$a_n = \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{n+1}}\, ds \quad , \quad b_n = \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{-n+1}}\, ds$$

QED.

Laurent series are often written as one sum:

$$\boxed{f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n \quad , \quad c_n = \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{n+1}}\, ds}$$

9.1.1 Properties

Note that if f is analytic in the region $R_1 < |z - z_0|$, the Laurent series representation is valid throughout that region.

Similarly, if f is analytic in $0 < |z - z_0| < R_2$, the Laurent series representation is valid throughout that deleted neighborhood.

Thus, if f is analytic for all $z \neq z_0$, then the Laurent series representation is valid for all $z \neq z_0$. This is useful for finding series representations of functions around a singularity, which Taylor series can't handle.

Laurent series reduces to Taylor series if f is analytic throughout the region (no singularity at center). To see this, note that if $|z| < R_2$, then

$$\begin{aligned}
a_n &= \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z)^{n+1}}\, ds = \frac{f^{(n)}(z_0)}{n!} && \text{(Cauchy integral formula)} \\
b_n &= \frac{1}{2\pi i} \int_{C} \frac{f(s)}{(s - z_0)^{-n+1}}\, ds \\
&= \frac{1}{2\pi i} \int_{C} f(s)(s - z_0)^{n-1}\, ds = 0 && \text{(Integrand analytic within } C\text{)}
\end{aligned}$$

9.1.2 Finding Laurent Series

Usually, the integral forms of a_n and b_n don't need to be computed to find the Laurent series representation. The Laurent series representation is unique (see section 72), so if we can find a series representation, it'll be a valid Laurent series.

Example: What's the Laurent series of $f(z) = e^z/(z-1)^3$ centered at $z = 1$?

Solution: Write $e^z = e \cdot e^{z-1}$. We know the Taylor series of e^{z-1} is

$$e^{z-1} = \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!}$$

Plug that into f to get

$$f(z) = e \cdot \frac{1}{(z-1)^3} \cdot \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} = \sum_{n=0}^{\infty} \frac{e}{n!}(z-1)^{n-3}$$

This works for all $z \neq 1$. The expression implies that $b_n = 0$ for all $n > 3$.

9.2 Uniform Convergence

For the next few lectures, the focus will shift from "what's the power series representation of this function?" to "what properties do functions defined by convergent power series have?" So the focus is shifting to power series in general.

Suppose the sequence of functions $f_1, f_2, \ldots$ and the function f are defined on some set A. We say the sequence **converges pointwise** to f on A if **for each** $z \in A$ and $\epsilon > 0$, there exists k such that

$$|f_n(z) - f(z)| < \epsilon \ , \ \ \forall n > k$$

I.e. for every $z \in A$,

$$\lim_{n\to\infty} f_n(z) = f(z)$$

A stronger requirement is **uniform convergence**. We say that the sequence **converges uniformly** to f on A if **for every** $\epsilon > 0$, there exists a k such that

$$|f_n(z) - f(z)| < \epsilon \ , \ \ \forall n > k \text{ and } \forall z \in A$$

For uniform convergence, the value of k is universal for all z. For pointwise convergence, the value of k could depend on the specific z.

Theorem: Suppose the sequence f_n converges uniformly to f on the set A. If f_n is continuous on A for all n, then f is continuous on A.

Proof: Suppose $z_0 \in A$. Let $\epsilon > 0$. Since f_n converges uniformly to f, there exists k such that $|f_k(z) - f(z)| < \epsilon/3$. Since f_k is continuous at z_0, there also exists $\delta > 0$ such that $|z - z_0| < \delta \implies |f_k(z) - f_k(z_0)| < \epsilon/3$.

Suppose $z \in A$ and $|z - z_0| < \delta$. Then by the triangle inequality,

$$|f(z) - f(z_0)| \leq \underbrace{|f(z) - f_k(z)|}_{<\epsilon/3} + \underbrace{|f_k(z) - f_k(z_0)|}_{<\epsilon/3} + \underbrace{|f_k(z_0) - f(z_0)|}_{<\epsilon/3}$$

$$|f(z) - f(z_0)| < \epsilon$$

We've shown that f is continuous on A.

Example: Let $f_n(z) = z^n$. This is continuous everywhere. But the function

$$f(z) = \begin{cases} 0 & z \neq 1 \\ 1 & z = 1 \end{cases}$$

isn't continuous at $z = 1$. Thus f_n can't converge uniformly to f.

Theorem: For each n, let f_n and f be functions on the set A. Let $0 < \sigma_n \in \mathbb{R}$. Suppose that for all n and $z \in A$, $|f_n(z) - f(z)| \leq \sigma_n$. If $\lim \sigma_n = 0$, then f_n converge uniformly to f on A.

That is, the "remainder" of $f_n(z) - f(z)$ must converge to 0 as $n \to \infty$.

Proof: Suppose $\lim \sigma_n = 0$ and $\epsilon > 0$. Then we can find a k such that $|\sigma_n| < \epsilon$ for all $n > k$. Then for all $z \in A$,

$$|f_n(z) - f(z)| \leq \sigma_n < \epsilon \ , \ \ \forall n > k$$

Thus f_n converges uniformly to f on A.

Uniform convergence can also be applied to series. Suppose that f_n is a function on the set A and there exists f such that for each $z \in A$, $f(z) = \sum_{n=0}^{\infty} f_n(z)$. We say that the series $\sum_{n=0}^{\infty} f_n$ **converges uniformly** to f on A if its **sequence of partial sums** converges uniformly to f on A.

Later, we'll show that if a power series converges, it converges to a continuous function.

10 Week 10 (Mar 16 – 20)

10.1 Continuity of Power Series

Our goal is to prove the following:

Functions defined by convergent power series are analytic.

We start by proving the continuity of power series within its disk of convergence.

If the series $\sum a_n(z - z_0)^n$ converges at some point other than z_0, the largest open disk $|z - z_0| < R$ on which the series converges is the **disk of convergence**. R is the **radius of convergence**.

If the series converges for all z, the disk of convergence is $\mathbb{C}$ and the radius of convergence is ∞.

Lemma 1: Suppose $\sum a_n(z - z_0)^n$ converges at the point z_1, where $|z_1 - z_0| = R_1 > 0$. Then the series is absolutely convergent at any z satisfying $|z - z_0| < R_1$.

Proof: Since $\sum a_n(z_1 - z_0)^n$ converges, $\lim a_n(z_1 - z_0)^n = 0$, and $\{a_n(z_1 - z_0)^n \mid n = 0, 1, 2, \dots\}$ is bounded. Let M satisfy $|a_n(z_1 - z_0)^n| < M$ for all n. Suppose z satisfies $|z - z_0| < |z_1 - z_0|$ (i.e. z is closer to z_0 than z_1).

Let $\rho = |z - z_0|/|z_1 - z_0| < 1$. For each n,

$$|a_n(z - z_0)^n| < M\rho^n$$

$\rho < 1$ implies that $\sum M\rho^n$ converges. By the comparison test, $\sum |a_n(z - z_0)^n|$ converges.

Lemma 2: Suppose $\sum a_n(z - z_0)^n$ has a radius of convergence R, and $|z_1 - z_0| = R_1 < R_0$. Then the series converges uniformly on $|z - z_0| < R_1$.

In other words, the power series convergences uniformly within its disk of convergence.

Proof: Let $S(z)$ denote the sum of the series and $S_k(z)$ the k-th partial sum. We want to show that S_k converges uniformly to S on $|z - z_0| < R_1$.

We can find a z_2 between z_1 and the disk's boundary, such that $|z_2 - z_0| > |z_1 - z_0|$ and the series converges at z_2. By **Lemma 1**, $\sum |a_n(z_1 - z_0)^n|$ converges.

For each k, denote the remainder ("tail") of the series as

$$\sigma_k = \sum_{n=0}^{\infty} |a_n(z_1 - z_0)^n| - \sum_{n=0}^{k-1} |a_n(z_1 - z_0)^n| = \sum_{n=k}^{\infty} |a_n(z_1 - z_0)^n|$$

Since the series converges, $\lim \sigma_k = 0$. For each k, find the difference of the partial sums. By the triangle inequality,

$$|S(z) - S_k(z)| = \left| \sum_{n=k}^{\infty} a_n(z - z_0)^n \right| \leq \sum_{n=k}^{\infty} |a_n(z - z_0)^n|$$

If $|z - z_0| \leq |z_1 - z_0| = R_1$, i.e. z is closer to z_0 than z_1, then

$$|S(z) - S_k(z)| \leq \sum_{n=k}^{\infty} |a_n(z - z_0)^n| \leq \sum_{n=k}^{\infty} |a_n(z_1 - z_0)^n| = \sigma_k$$

QED.

Theorem: A power series $S(z) = \sum a_n(z - z_0)^n$ represents a continuous function at each point inside its disk of convergence.

Proof: Suppose w is within the disk of convergence. Find a z_1 such that $|z_1 - z_0| > |w - z_0|$. By **Lemma 2**, $\sum a_n(z - z_0)^n$ converges uniformly to S on the disk $|z - z_0| < |z_1 - z_0|$. Thus the sequence of partial sums S_k converge uniformly to S within the disk.

Each S_k is just a polynomial, so it's continuous. Continuous functions converging uniformly must converge to a continuous function, so S is continuous on this disk. And w is on this disk, so S is continuous at w. This generalizes for all w in general within the disk.

The above theorem generalizes to Laurent series as well. Consider the series

$$\sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} b_n(z-z_0)^{-n}$$

If the above converges for the annulus $R_1 < |z-z_0| < R_2$, then both series converge uniformly on any closed annulus concentric to and interior to $R_1 < |z-z_0| < R_2$.

10.2 Analyticity of Power Series

Lemma 1: Let $S(z) = \sum a_n(z-z_0)^n$ and let C be any contour lying within the disk of convergence of the power series. If g is continuous on C, then

$$\int_C g(z)S(z)\ dz = \sum_{n=0}^{\infty} a_n \int_C g(z)(z-z_0)^n\ dz$$

Proof: Let the k-th remainder ("tail") of the series be $\rho_k(z) = \sum_{n=k}^{\infty} a_n(z-z_0)^n$. Then

$$g(z)S(z) = \sum_{n=0}^{k-1} a_n g(z)(z-z_0)^n + g(z)\rho_k(z)$$

Integrate each side. Note that S, g are continuous on C, so $\int_C g(z)S(z)\ dz$ exists. Additionally, the finite sum $\sum_{n=0}^{k-1} a_n g(z)(z-z_0)^n$ is continuous on C, so $\int_C \sum_{n=0}^{k-1} a_n g(z)(z-z_0)^n\ dz$ exists. Therefore, $\int_C g(z)\rho_k(z)\ dz$ exists.

$$\int_C g(z)S(z)\ dz = \sum_{n=0}^{k-1} a_n \int_C g(z)(z-z_0)^n\ dz + \int_C g(z)\rho_k(z)\ dz$$

It suffices to show that the last term goes to 0 as $k \to \infty$. Let $M = \max\{|g(z)| \mid z \in \mathbb{C}\}$, and L be the length of C. There exists a closed disk $|z-z_0| \le R$ lying inside the disk of convergence and containing C. On that region, the series converges uniformly. Let $\epsilon > 0$. There exists k' such that for all $k > k'$ and $|z-z_0| \le R$,

$$|\rho_k(z)| < \frac{\epsilon}{ML}$$
$$|g(z)\rho_k(z)| < \frac{M\epsilon}{ML} = \frac{\epsilon}{L}$$
$$\left|\int_C g(z)\rho_k(z)\ dz\right| < \frac{\epsilon}{L}\cdot L = \epsilon$$

Thus $\lim_{k\to\infty} \int_C g(z)\rho_k(z)\ dz = 0$, and

$$\int_C g(z)S(z)\ dz = \sum_{n=0}^{\infty} a_n \int_C g(z)(z-z_0)^n\ dz$$

Theorem: A power series $S(z) = \sum a_n(z-z_0)^n$ is analytic at each point inside its disk of convergence.

Proof: Let D be the open disk of convergence, and let $g(z) = 1$. If C is a closed contour within D, then by **Lemma 1**,

$$\int_C S(z)\ dz = \sum_{n=0}^{\infty} a_n \int_C (z-z_0)^n\ dz$$

But $(z-z_0)^n$ is analytic throughout the domain D, so for any closed curve C,

$$\int_C (z-z_0)^n\ dz = 0$$

Then $\int_C S(z)\ dz = 0$. Since S is continuous in D, and the integral of S along any closed contour in D is zero, S must be analytic throughout D by Morera's theorem.

We've shown that a power series is both continuous and analytic within its disk of convergence. This can be used to, for example, differentiate power series term-by-term.

Theorem: If $S(z) = \sum a_n(z-z_0)^n$, then within its disk of convergence,

$$S'(z) = \sum_{n=1}^{\infty} na_n(z-z_0)^{n-1}$$

Proof: Let z be a point inside the disk of convergence. Define g as

$$g(w) = \frac{1}{2\pi i} \cdot \frac{1}{(w-z)^2}$$

Let C be a simple closed contour contained in the disk of convergence **going around** z. By the Cauchy integral formula,

$$\begin{aligned} S'(z) &= \frac{1}{2\pi i}\int_C \frac{S(w)}{(w-z)^2}\ dw \\ &= \int_C g(w)S(w)\ dw \\ &= \sum_{n=0}^{\infty} a_n \int_C g(w)(w-z_0)^n\ dw \end{aligned}$$

Substitute for $g(w)$ to get

$$\begin{aligned} S'(z) &= \sum_{n=0}^{\infty} a_n \cdot \frac{1}{2\pi i}\int_C \frac{(w-z_0)^n}{(w-z_0)^2}\ dw \\ &= \sum_{n=0}^{\infty} a_n \frac{d}{dz}\left[(z-z_0)^n\right] && \text{(Cauchy integral formula)} \\ &= \sum_{n=1}^{\infty} na_n(z-z_0)^{n-1} \end{aligned}$$

Side note: it's not just addition and subtraction. Multiplication of power series term-by-term is valid within their disks of convergence. E.g. Leibniz's rule:

$$\left(f(z)g(z)\right)^n = \sum_{j=0}^{n} \binom{n}{j} f(z)^j g(z)^{n-j}$$

Division works too, but that gets complicated.

11 Spring Break (Mar 23 – 27)

12 Week 12 (Mar 30 – Apr 3)

12.1 Isolated Singularities and Residues

We want to understand how singular points within a closed contour affect the integral around a contour.

Suppose f is not analytic at z_0 but is analytic in some deleted neighborhood of z_0, $0 < |z - z_0| < \varepsilon$. Then z_0 is an **isolated singular point** of f. For example, $z = 0$ is an isolated singularity of $f(z) = 1/z$.

Some singularities aren't isolated. E.g. consider

$$f(z) = \frac{1}{e^{1/z} - 1}$$

This isn't analytic at $z = 0$ or $z_n = \frac{1}{2n\pi i}$. But note that $\lim_{n\to\infty} z_n = 0$, so at every neighborhood of 0, we have infinitely many singular points other than 0.

Let f be analytic in some disk $0 < |z - z_0| < R$ for some $R > 0$. Let z_0 be an isolated singular point of f, and C a simple closed contour around z_0 lying within $0 < |z - z_0| < R$. Then

$$\text{Res}_{z=z_0} f = \frac{1}{2\pi i} \int_C f(z)\, dz$$

is called the **residue of f at** z_0, denoted $\text{Res}_{z=z_0} f$.

*By the principle of deformation of paths, the specific contour doesn't matter as long as it's on the region $0 < |z - z_0| < R$.

**E.g. For $f(z) = 1/z$ with a singularity at $z = 0$, the residue is

$$\text{Res}_{z=0} f = \frac{1}{2\pi i} \int_C \frac{1}{z}\, dz = 1$$

Finding the residue is easy if we have a valid Laurent series representation in $0 < |z - z_0| < R$. If z_0 is a singular point of f and

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n\ ,\ \ 0 < |z - z_0| < \varepsilon$$

Then the residue is

$$\text{Res}_{z=z_0} f = c_{-1}$$

Proof: For any n, the coefficient of the Laurent series was given by

$$c_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-z_0)^{n+1}}\, dz$$

12.2 Residue Theorems

Theorem (Cauchy's Residue Theorem): Let C be a simple closed contour, and suppose f is analytic inside and on C, except at the singular points $z_1, z_2, \ldots, z_n \in C$. Then the integral of f along C (oriented counterclockwise) is

$$\int_C f(z)\, dz = 2\pi i \sum_{k=1}^{n} \text{Res}_{z=z_k} f$$

Proof: There's a finite number of z_k, so each z_k is isolated. For each k, there exists a deleted neighborhood of z_k on which f is analytic. Let C_k be a simple closed contour around each z_k. Then f is analytic in every region interior to C but exterior to C_k. By the principle of deformation of paths,

$$\begin{aligned}\int_C f(z)\, dz &= \sum_{k=1}^{n} \int_{C_k} f(z)\, dz \\ &= 2\pi i \sum_{k=1}^{n} \text{Res}_{z=z_k} f\end{aligned}$$

Theorem: Suppose f is analytic everywhere except a finite number of points inside a simple closed contour C, oriented counterclockwise. Let $g(z) = \frac{1}{z^2} f(1/z)$. Then

$$\int_C f(z)\, dz = 2\pi i\, \text{Res}_{z=0} g$$

This theorem allows us to get away with computing only 1 residue, even if there are multiple.

Proof: Find some R_1 such that C is contained in the disk $|z| < R_1$. Let $R_0 > R_1$, and let C_0 be the circle $|z| = R_0$. Since f is analytic everywhere on $|z| > R_1$, it has a Laurent series representation on C_0:

$$\begin{aligned} f(z) &= \sum_{n=-\infty}^{\infty} c_n z^n\ , \quad |z| > R_1 \\ c_n &= \frac{1}{2\pi i} \int_{C_0} \frac{f(z)}{z^{n+1}}\, dz \end{aligned}$$

Let $n = -1$. Use the principle of deformation of paths to deform C_0 into C.

$$2\pi i\, c_{-1} = \int_{C_0} f(z)\, dz = \int_C f(z)\, dz$$

It suffices to show that c_{-1} is the residue of g at $z = 0$. We can find $g(z)$ to be

$$g(z) = z^{-2} f(1/z) = \sum_{n=-\infty}^{\infty} c_n z^{-n-2} \ , \ \ 0 < |z| < R_1$$

This is valid Laurent series for g in the given domain. Thus, the residue of g at $z = 0$ is the coefficient of the $1/z$ term, which is c_{-1}.

12.3 Classification of Isolated Singularities

If z_0 is an isolated singular point of f, then f has a Laurent series representation in some neighborhood of z_0.

$$f(z) = \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{non-principal part}} + \underbrace{\sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}}_{\text{principal part}} \ , \ \ |z - z_0| < \varepsilon$$

We call the second term the principal part of f at z_0. Note that the non-principal part converges,

$$\lim_{z \to z_0} \sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0$$

So the behavior of f close to z_0 is **determined by the principal part** of the Laurent series.

Suppose that in the principal part of f at z_0, there exists m such that

$$b_m \neq 0 \ \ \textit{and} \ \ b_j = 0 \ , \ \ \forall j > m$$

Then z_0 is called a **pole of order** m **of** f. I.e. the largest power of $1/(z - z_0)$ in the Laurent representation.

- If there are infinitely many m such that $b_m \neq 0$, then z_0 is called an **essential singularity** of f.
- If $b_m = 0$ for all m, then z_0 is called a **removable singularity** of f.
- Else, there's finitely many m such that $b_m \neq 0$ and $b_j = 0$ for all $j > m$. So z_0 is just called a **pole of order** m.

For example, consider $f(z) = e^{1/z}$. This has one isolated singularity at $z_0 = 0$. The Laurent series representation is

$$f(z) = 1 + \frac{1}{z} + \frac{1}{2!\, z^2} + \cdots$$

There are infinitely many powers of $1/z$, so 0 is an **essential singularity** of f.

Theorem: The isolated singularity z_0 of f is a pole of order m iff. there exists $\varphi(z)$, analytic in a neighborhood of z_0, such that $\varphi(z_0) \neq 0$ and

$$f(z) = \frac{\varphi(z)}{(z - z_0)^m}$$

for all $z \neq z_0$ in that neighborhood.

Proof: Power series stuff.

Corollary 1: If f has a pole of order m and $f(z) = \varphi(z)/(z - z_0)^m$, where φ is analytic and nonzero at z_0, then

$$\text{Res}_{z=z_0} f = \frac{\varphi^{(m-1)}(z_0)}{(m-1)!}$$

Proof: For $z \in C$, we have that $f(z) = \varphi(z)/(z - z_0)^m$. Using residue theorems and the generalized Cauchy integral formula,

$$\begin{aligned}\text{Res}_{z=z_0} f = \frac{1}{2\pi i}\int_C f(z)\,dz &= \frac{1}{2\pi i}\int_C \frac{\varphi(z)}{(z-z_0)^m}\,dz \\ &= \frac{\varphi^{(m-1)}(z_0)}{(m-1)!}\end{aligned}$$

Corollary 2: If f has a pole of order m at z_0, then

$$\lim_{z \to z_0} f(z) = \infty$$

Proof: Start off with $f(z) = \varphi(z)/(z - z_0)^m$ for $0 < |z - z_0| < \varepsilon$. Here, φ is analytic in $|z - z_0| < \varepsilon$, and $\varphi(z_0) \neq 0$. This is just from the theorem.

Now, φ is continuous and nonzero at z_0. There must exist a smaller deleted neighborhood D of z_0 in which $\varphi(z)$ is nonzero. Hence, $f(z)$ is nonzero in D, and $1/f(z)$ is defined.

$$\lim_{z \to z_0} \frac{1}{f(z)} = \lim_{z \to z_0} \frac{(z - z_0)^m}{\varphi(z)} = \frac{0}{\varphi(z_0)} = 0$$

Later we'll see that: $\lim_{z \to z_0} f(z) = \infty$ **only when** z_0 **is a pole**.

13 Week 13 (Apr 6 – Apr 10)

13.1 Behavior of a Function Near Singularities

Near a removable singularity, a function looks "well-behaved." Near an essential singularity, a function is "wild."

13.1.1 Near Removable Singularities

Theorem: If z_0 is a removable singularity of f, there exists a function g, analytic throughout some neighborhood D of z_0, such that $f(z) = g(z)$ for all $z \in D \setminus \{z_0\}$.

Proof: Since z_0 is removable, there exists a neighborhood D of z_0 such that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n\ , \quad \forall z \in D \setminus \{z_0\}$$

This converges for all $z \in D$. Let $g(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ for all $z \in D$. Then $f(z) = g(z)$ for all $z \in D \setminus \{z_0\}$. Note that g is analytic in D because it's the sum of a convergent power series. QED.

Corollary: If f has a removable singularity at z_0, then $\lim_{z \to z_0} f(z)$ exists, and

$$\phi(z) = \begin{cases} f(z) & z \neq z_0 \\ \lim_{z \to z_0} f(z) & z = z_0 \end{cases}$$

is analytic at z_0.

Proof: Let g be the function given in the previous theorem. Then $\lim_{z \to z_0} f(z) = g(z_0)$. Note that

$$\begin{cases} \phi(z) = g(z) = f(z) & z \neq z_0 \\ \phi(z_0) = \lim_{z \to z_0} f(z) = g(z_0) & z = z_0 \end{cases}$$

So $\phi(z) = g(z)$, which is analytic.

The following theorems strongly imply that z_0 is a removable singularity of f iff. f is **bounded in some deleted neighborhood** of z_0. The exact proof is a little more complicated.

13.1.2 Near Essential Singularities

Theorem (Casorati-Weierstrass): Let z_0 be an essential singularity of f, and let D be any neighborhood of z_0. If $w_0 \in \mathbb{C}$ and $\varepsilon > 0$, then there exists $z \in D$ such that

$$|f(z) - w_0| < \varepsilon$$

I.e. In every neighborhood of an essential singularity z_0, the function takes values arbitrarily close to any complex number.

Proof: We use proof by contradiction. Let $D^* = D \setminus \{z_0\}$. Assume that z_0 is an essential singularity of f, and there exists $\varepsilon > 0$ and $w_0 \in \mathbb{C}$ such that

$$|f(z) - w_0| \geq \varepsilon \ , \quad \forall z \in D^*$$

Define the function

$$g(z) = \frac{1}{f(z) - w_0} \ , \quad z \in D^*$$

Note that since $|f(z) - w_0| \geq \varepsilon$, we must have $|g(z)| \leq 1/\varepsilon$ for $z \in D^*$. That is, $g(z)$ is bounded in some deleted neighborhood of z_0, thus the singularity z_0 must be removable and the limit $\lim_{z \to z_0} g(z)$ exists. Let $g(z_0) = \lim_{z \to z_0} g(z)$.

Case 1: Suppose $g(z_0) \neq 0$. Then

$$\frac{1}{g(z_0)} = \lim_{z \to z_0} \frac{1}{1/(f(z) - w_0)} = \lim_{z \to z_0} f(z) - w_0$$

For this to exist, $\lim_{z \to z_0} f(z)$ must exist, so f must be bounded in D^*. Thus, z_0 must be a removable singularity of f. But we were assuming that z_0 is an essential singularity of f, so this is a contradiction.

Case 2: Suppose $g(z_0) = 0$. Write $g(z)$ in general as

$$g(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots, \quad z \in D$$

Since $g(z_0) = 0$, we have $a_0 = 0$. Let a_m be the first nonzero coefficient. Define $\phi(z)$ such that

$$\phi(z) = a_m + a_{m+1}(z - z_0) + \cdots$$
$$g(z) = (z - z_0)^m(a_m + a_{m+1}(z - z_0) + \cdots) = (z - z_0)^m\phi(z)$$

For all $z \in D^*$, we have $\phi(z) = g(z)/(z - z_0)^m$. Note that $\phi(z_0) = a_m \neq 0$, and $1/\phi$ is analytic in D. Write

$$\frac{1}{\phi(z)} = c_0 + c_1(z - z_0) + c_2(z - z_0)^2 + \cdots, \quad z \in D$$

Then

$$\begin{aligned} f(z) &= \frac{1}{g(z)} + w_0 = \frac{1}{(z - z_0)^m} \cdot \frac{1}{\phi(z)} + w_0 \\ &= \frac{1}{(z - z_0)^m}\left(c_0 + c_1(z - z_0) + c_2(z - z_0)^2 + \cdots\right) + w_0 \end{aligned}$$

Thus z_0 is a pole of order m of f. But this is a contradiction, since we assumed that z_0 was an essential singularity of f.

An even stronger result is Picard's theorem, which says that in every neighborhood of an essential singularity, a function takes on **every finite value** (with one possible exception) an **infinite amount of times**.

Proof: Too hard.

Example: Let $f(z) = e^{1/z}$. Suppose $w = re^{i\theta}$, and $for n = \mathbb{Z}_+$, let

$$z_n = \frac{1}{\ln r + i(\theta + 2\pi n)}$$

Since $\lim_{n\to\infty} z_n = 0$, there are infinitely many z_n in any neighborhood of 0. Note that

$$f(z_n) = e^{1/z_n} = re^{i\theta} = w$$

So in a neighborhood of 0, we can manipulate n so that every value of w appears an infinitely many number of times.

13.1.3 Summary of Results

Suppose z_0 is an isolated singularity of f.

1. If z_0 is removable, $\lim_{z\to z_0} f(z)$ exists and is a finite number.
2. If z_0 is a pole, $\lim_{z\to z_0} f(z) = \infty$.
3. If z_0 is an essential singularity, $\lim_{z\to z_0} f(z)$ does not exist, neither as a finite number nor as ∞.

(1) and (2) have been proven. (3) follows from Casorati-Weierstrass.

13.2 Zeros and Poles

Suppose f is analytic at z_0 and $f(z_0) = 0$. Then z_0 is a **zero of order** m if $f'(z_0) = f''(z_0) = \cdots = f^{(m-1)}(z_0) = 0$, but $f^{(m)}(z_0) \neq 0$.

Lemma: Suppose f is analytic at z_0. For each positive m, z_0 is a zero of order m for f iff. there exists g (analytic and nonzero at z_0) such that

$$f(z) = (z - z_0)^m g(z)$$

for all z in a neighborhood of z_0.

Example: z_0 is a zero of order m for $(z - z_0)^m$, and it's a pole of order m for $1/(z - z_0)^m$.

Theorem: Suppose $p(z)$ and $q(z)$ are analytic at z_0, and $p(z_0) \neq 0$. If $q(z)$ has a zero of order m at z_0, then $p(z)/q(z)$ has a pole of order m at z_0.

Proof: Write $q(z) = (z - z_0)^m g(z)$, where g is analytic at z_0 and $g(z_0) \neq 0$. By continuity of g at z_0,

1. $g(z) \neq 0$ throughout a neighborhood of z_0.
2. $q(z) = (z - z_0)^m g(z) \neq 0$ in some deleted neighborhood of z_0.

Thus z_0 is the only singularity of $p(z)/q(z)$ in this neighborhood. I.e. z_0 is an isolated singularity of $p(z)/q(z)$. Now write

$$\frac{p(z)}{q(z)} = \frac{p(z)/g(z)}{(z - z_0)^m}$$

Since $g(z_0) \neq 0$, $p(z)/g(z)$ is analytic in a neighborhood of z_0. I.e.

$$\frac{p(z)}{g(z)} = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

$$\frac{p(z)}{q(z)} = \frac{c_0}{(z - z_0)^m} + \frac{c_1}{(z - z_0)^{m-1}} + \cdots$$

Note that $p(z_0) \neq 0$ (assumption in theorem) implies that $c_0 \neq 0$. Thus z_0 is a pole of $p(z)/q(z)$ of order m.

This provides a simple way to find residues.

Theorem: Suppose $p(z)$ and $q(z)$ are analytic at z_0, $p(z_0) \neq 0$, $q(z_0) = 0$, and $q'(z_0) \neq 0$. Note that z_0 is a zero of order 1 of $q(z)$. Then z_0 is a simple pole of $p(z)/q(z)$, i.e. a pole of order 1. Then

$$\text{Res}_{z=z_0} \frac{p}{q} = \frac{p(z_0)}{q'(z_0)}$$

13.3 Uniqueness of Analytic Functions

Theorem: Suppose f, g are analytic on the domain D, $S \subset D$, and $f(s) = g(s)$ for all $s \in S$. If S has an accumulation point in D, then $g(z) = g(z)$ for all $z \in D$.

Lemma 1: Let f be analytic in the neighborhood N of z_0, and suppose that in every deleted neighborhood of z_0, there exists z such that $f(z) = 0$. Then $f(z) = 0$ for all z in N.

Lemma 2: Suppose f is analytic in a domain D, $S \subset D$, and $f(s) = 0$ for all $s \in S$. If S has an accumulation point $z_0 \in D$, then $f(z) = 0$ for all $z \in D$.

Proof of Theorem: Follows immediately from Lemmas using $h(z) = f(z) - g(z)$. For more details, see Lecture 45.

There's a corollary saying

Corollary: If $f(x)$ is a real-valued function of a real variable and D is any domain in the complex plane which contains a real number, then there is at most one function ϕ such that ϕ is analytic on D and $\phi(x) = f(x)$ for all $x \in \mathbb{R}$.

What this says is, **extensions of real functions to complex analytic functions are unique**. Thus we're justified in our choice of the complex log, the complex exponential, complex trig functions, etc.

13.4 Summary

Since it's been a long time, here's a summary of the key equations/concepts we have so far.

Cauchy-Riemann Equations: If $f(z) = u(x, y) + iv(x, y)$, then $u_x = v_y$ and $u_y = -v_x$.

Cauchy-Goursat theorem: If f is analytic on and within a simple closed contour C, then

$$\int_C f(z)\, dz = 0$$

Principle of Deformation of Paths: If C_1, C_2 are two non-intersecting simple closed contours, C_2 is inside C_1, and f is analytic on the contours and at all points between them, then

$$\int_{C_1} f(z)\, dz = \int_{C_2} f(z)\, dz$$

Generalized Cauchy Integral Formula: If f is analytic on and within the simple closed contour C, then

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz$$

Power Series Representations: If f is analytic at z_0, then in some neighborhood of z_0,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n\ , \quad a_n = \frac{f^{(n)}(z_0)}{n!}$$

Laurent Series Representations: If f is analytic in the annulus $R_1 < |z - z_0| < R_2$, and C is a simple closed contour lying in the annulus, then in this domain,

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n$$
$$c_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz$$

14 Week 14 (Apr 13 – Apr 17)

14.1 Improper Integrals

If $f(x)$ converges for all x, the improper integral over the entire interval is

$$\int_{-\infty}^{+\infty} f(x)\, dx = \lim_{R_1 \to \infty} \int_{-R_1}^{0} f(x)\, dx + \lim_{R_2 \to \infty} \int_{0}^{R_2} f(x)\, dx \tag{1}$$

When both limits exist, we say the integral on the left **converges** to the sum on the right.

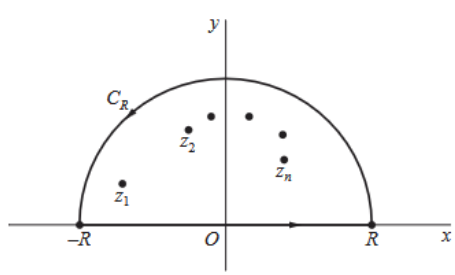
Alternatively, the **Cauchy principal value** of the integral is

$$\text{P.V.} \int_{-\infty}^{+\infty} f(x)\, dx = \lim_{R \to \infty} \int_{-R}^{R} f(x)\, dx \tag{2}$$

If integral (1) converges, its Cauchy P.V. exists. But it isn't true that integral (1) converges just because its Cauchy P.V. exists. The exception is if f is even, $f(-x) = f(x)$. If this is the case, and the Cauchy PV exists, then integral (1) exists.

$$\int_{-\infty}^{+\infty} f(x)\, dx = \text{P.V.} \int_{-\infty}^{+\infty} f(x)\, dx$$

There's a clever way to evaluate improper integrals of rational functions, by using sum of residues. Let $f(z) = p(z)/q(z)$, where $q(z)$ has zeros at $z_1, z_2, \ldots, z_n$. Draw a contour with two parts, $-R \le x \le R$ and C_R.



Half-circle contour (Brown & Churchill 261)

From Cauchy's residue theorem, we have

$$\int_{-R}^{R} f(x)\, dx + \int_{C_R} f(z)\, dz = 2\pi i \sum_{k=1}^{n} \text{Res}_{z=z_k} f(z)$$

If

$$\lim_{R\to\infty}\int_{C_R} f(z)\,dz = 0$$

Then

$$\text{P.V.}\int_{-\infty}^{+\infty} f(x)\,dx = 2\pi i\sum_{k=1}^{n}\text{Res}_{z=z_k} f(z)$$

In the case that f is even, the improper integral is equal to its principal value, so

$$\int_{-\infty}^{+\infty} f(x)\,dx = 2\pi i\sum_{k=1}^{n}\text{Res}_{z=z_k} f(z) \tag{1}$$

$$\int_{0}^{\infty} f(x)\,dx = \pi i\sum_{k=1}^{n}\text{Res}_{z=z_k} f(z) \tag{2}$$

Example: Consider the following integral

$$\int_0^\infty \frac{1}{x^6+1}\,dx$$

It's hard to evaluate classically. But consider the function $f(z) = 1/(z^6+1)$. The zeros on the upper half-circle are $c_0 = e^{i\pi/6}$, $c_1 = i$, and $c_2 = e^{i5\pi/6}$. The residues are

$$\text{Res}_{z=z_k}\frac{1}{z^6+1} = \frac{1}{6c_k^5} = \frac{c_k}{6c_k^6} = -\frac{c_k}{6}$$

Thus the contour integral is

$$\begin{aligned}\int_{-R}^{R} f(x)\,dx + \int_{C_R} f(z)\,dz &= 2\pi i\sum_{k=0}^{2}\text{Res}_{z=c_k}\\ &= \frac{2\pi}{3}\end{aligned}$$

All that's left is to show that the 2nd integral on the LHS goes to zero as $R\to\infty$. Note that if $R>1$, $|z^6+1| \geq ||z|^6 - 1| = R^6 - 1$. Therefore,

$$|f(z)| \leq \frac{1}{R^6-1}$$

$$\left|\int_{C_R} f(z)\,dz\right| \leq \frac{\pi R}{R^6-1}\ ,\ \text{ goes to } 0 \text{ as } R\to\infty$$

Therefore our integral is

$$\int_0^\infty \frac{1}{x^6+1}\,dx = \frac{1}{2}\lim_{R\to\infty}\int_{-R}^{R} f(x)\,dx = \frac{\pi}{3}$$

14.2 Improper Integrals from Fourier Analysis

Consider integrals of the form below:

$$\int_{-\infty}^{+\infty} f(x)\sin(ax)\ dx \quad \text{or} \quad \int_{-\infty}^{+\infty} f(x)\cos(ax)\ dx$$

These aren't easily integrable with the previous method. $|\sin(az)|^2 = \sin^2(ax) + \sinh^2(ay)$ and $|\cos(az)|^2 = \cos^2(ax) + \sinh^2(ay)$, both of which go to infinity as $y \to \infty$. Instead, we use a clever trick, considering only the real/imaginary parts of e^{iaz}.

Example: We try to evaluate the integral

$$\int_0^{\infty} \frac{\cos(2x)}{(x^2+4)^2}\ dx$$

Let $f(z) = 1/(z^2+4)^2$ and introduce e^{i2z} into the equation. The half-circle contour from before, with $R > 2$, is

$$\int_{-R}^{R} \frac{e^{i2x}}{(x^2+4)^2}\ dx = 2\pi i B - \int_{C_R} f(z)e^{i2z}\ dz$$
$$B = \text{Res}_{z=2i}[f(z)e^{i2z}]$$

It turns out that the residue $B = 5/(32e^4 i)$. Next we equate the real parts of the equation

$$\int_{-R}^{R} \frac{\cos(2x)}{(x^2+4)^2}\ dx = \frac{5\pi}{16e^4} - \text{Re}\int_{C_R} f(z)e^{i2z}\ dz$$

We show that the 2nd integral on the RHS goes to zero as $R \to \infty$ and conclude.

14.2.1 Jordan's Lemma

To evaluate integrals of the form above, **Jordan's lemma** is helpful.

Theorem (Jordan's Lemma): Suppose that

1. A function $f(z)$ is analytic at all points in the upper half plane $y \geq 0$ that are exterior to a circle $|z| = R_0$.
2. C_R denotes a semicircle $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, where $R > R_0$.
3. For all points z on C_R, there exists a positive constant M_R such that

$$|f(z)| \leq M_R \ \text{ and } \ \lim_{R\to\infty} M_R = 0$$

Then for every positive a,

$$\lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}\ dz = 0$$

14.3 Definite Integrals Involving Sines and Cosines

Consider the definite integral

$$\int_0^{2\pi} F(\sin\theta, \cos\theta)\, d\theta$$

Use the parameterization $z = e^{i\theta}$ and make the substitutions (from representations of sine and cosine)

$$\sin\theta = \frac{z - z^{-1}}{2i}\ , \quad \cos\theta = \frac{z + z^{-1}}{2}\ , \quad d\theta = \frac{dz}{iz}$$

Then the integral becomes a contour integral around C (circle of radius 1):

$$\int_0^{2\pi} F(\sin\theta, \cos\theta)\, d\theta = \int_C F\left(\frac{z - z^{-1}}{2i}, \frac{z + z^{-1}}{2}\right) \frac{dz}{iz}$$

This is usually applicable when we have a $\sin\theta$ or $\cos\theta$ in the denominator of the integrand, and the numerator is simple.

Example: We want to integrate the following.

$$\int_0^{2\pi} \frac{1}{1 + a\sin\theta}\, d\theta$$

Make the substitutions to get (where C is the circle $|z| = 1$)

$$\int_0^{2\pi} \frac{1}{1 + a\sin\theta}\, d\theta = \int_C \frac{2/a}{z^2 + (2i/a)z - 1}\, dz$$

Use Cauchy's residue theorem and simple residues of poles to get

$$\int_C \frac{2/a}{z^2 + (2i/a)z - 1}\, dz = 2\pi i \left(\frac{1}{i\sqrt{1 - a^2}}\right) = \frac{2\pi}{\sqrt{1 - a^2}}$$

QED.